PRD: Hackathon & Case Competition Management Platform

22 Oct 2026 · @Neelanshu Karn s Team

0. Document control

This PRD is the primary specification for the first development cycle (V1) and the pilot at one real college hackathon or case competition. It recommends a smaller V1 than the full brief: native file storage, one-way Google Sheets push, and no automatic score normalisation.

Item	Value
Document version	0.1 (draft for team review)
Status	Draft; decisions marked "Recommendation" need team sign-off
Product	Hackathon & Case Competition Management Platform
Audience	Product, design, engineering, QA, pilot organisers
Scope	V1 (pilot release) plus clearly marked future items
Technology stack	Not specified; this PRD is stack-agnostic

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0.2 Statement labels

Label	Meaning
Confirmed	Stated in the product brief; not open for debate in V1
Recommendation	The author's proposed decision; needs team sign-off
Assumption	Believed true but unverified; must be validated in pilot or with organisers
Open question	Unresolved; tracked in Section 17

0.3 Priority definitions

Priority	Meaning

P0	Required for the first usable release; the pilot cannot run without it
P1	Important; may be simplified or delivered after P0 is stable, but planned for V1
P2	Future enhancement; not built in V1

0.4 Working assumptions

ID	Assumption	Impact if wrong
AS-01	A single event has at most 1,000 participants, 250 teams, 50 judges and 30 staff	Performance targets and list UX need revisiting
AS-02	Fewer than 10 events run on the platform at the same time during the pilot period	Capacity planning changes
AS-03	All users have an email address they can access during the event	Authentication design changes
AS-04	Judges evaluate on laptops; staff and participants may use phones	Mobile layouts become mandatory for staff and participant screens
AS-05	Venue internet can be unreliable but is not absent for long periods	An offline mode would be needed
AS-06	One organising team runs one event; no cross-organisation administration is needed	Multi-tenancy would be required

AS-07	Event-day judging is mostly asynchronous review of submissions, sometimes alongside live pitches	Live-pitch scheduling would be needed
AS-08	Organisers can provide one Google account to connect integrations	Integration ownership model changes
AS-09	Times are entered and displayed in the event's configured time zone (default Asia/Kolkata for the pilot)	Deadline display and enforcement logic change
AS-10	No payments are collected through the platform in V1	Payment module needed

0.5 Key V1 decisions at a glance

Decision	Recommendation	Section
Source of truth	Platform database for all records	6-F, 9
Google Sheets	One-way push (platform to sheet) of platform-owned tabs, plus CSV export as fallback	6-F
File storage	Native platform storage is canonical; Google Drive is an optional one-way archive mirror (P1)	6-G
Scoring	Normalise each criterion to 0–1, weights must total exactly 100, team score is mean of submitted judge scores	6-E

Ties	Detected and flagged; resolved manually by an organiser with a recorded reason	6-E
Roles	One role per user per event; a user can hold different roles in different events	8
Anonymous judging	Not in V1	6-C
Rounds and tracks	Every event has at least one round; tracks are optional	5, 9

1. Executive summary

V1 lets one college organising team run a hackathon or case competition from registration to certificates in one platform, with Google Sheets and Drive as supporting integrations rather than the system of record.

Product overview. A web platform with four role-specific interfaces (Core Admin, Participant/Team, Judge, Helping Staff) inside one application. Each user sees only the screens and records their role in that event allows.

Problem. Organisers stitch events together from Google Forms, Sheets, Docs, Drive, WhatsApp and separate scoring spreadsheets. Records are duplicated, submissions get attached to the wrong team, judges score inconsistently, results are calculated by hand, and certificates and feedback are separate manual jobs.

Solution. One canonical database holding the event, its teams, submissions, evaluations, results, feedback and certificates. The platform pushes participant and team data to a connected Google Sheet for organisers who work in spreadsheets, and can mirror submission files to Google Drive for archiving.

Primary users. College hackathon and case competition organisers (student committees and faculty coordinators). Secondary users are participants, team leaders, judges and volunteers.

Core value proposition.

- Organisers: one place to see who registered, who submitted, who has judged whom, and what the results are.
- Participants: one link for rules, team status, submission status, results and certificates.
- Judges: a focused queue of assigned teams with the rubric next to the submission.
- Staff: a task list and a fast check-in screen without access to scores.

MVP scope. Event setup, registration and teams, editable guidebook with PDF export, structured submissions with deadline enforcement, judge assignment, generic and custom rubrics, weighted scoring, result review and publication, basic staff tasks and check-in, participant and judge feedback, certificate generation and release, CSV export, one-way Sheets push, and essential audit history.

Differentiators (for this segment).

1. The scoring formula is defined, validated and auditable, so results can be defended to participants.
2. Submissions are bound to a team identity and locked at the deadline, removing the "which file is final?" problem.
3. Role separation is enforced on the server: volunteers and judges cannot see what they should not.
4. Integrations report their real status; a failed sync never shows as successful.

Target launch outcome. Run one real event end to end on the platform with no fallback to spreadsheets for scoring or results, every submission linked to the correct team, results published from the platform, and certificates downloaded by participants. Pilot measures are defined in Section 12.

2. Problem definition and user research hypotheses

The problems below come from the product brief and the team's own event experience; none are backed by formal research yet. Each is tagged **Established** (stated in the brief as a known operational problem) or **Hypothesis** (needs validation with organisers).

2.1 Organiser problems

ID	Problem	Current workflow	Why current tools fall short	Proposed improvement	Expected measurable outcome	Status

OP-1	Duplicated, inconsistent participant records	Google Form responses copied into several Sheets; edits made in one copy only	Forms has no team concept; no single record per person	One participant record per user per event; team membership as a relationship	Count of manual record corrections per event	Established
OP-2	Team formation is messy	Leaders submit member emails in a form; organisers reconcile by hand	No validation of team size or one-team-per-person	Join codes, size limits and one-team rule enforced by the system	Teams violating size or uniqueness rules at registration close (target 0)	Established
OP-3	Submissions misplaced or mis-attributed	Teams upload to a shared Drive folder or email links	Files are named by participants; no binding to team ID	Submission form bound to the logged-in team, versioned, locked at deadline	% submissions linked to correct team (target 100%)	Established
OP-4	Missing submissions hard to track	Organiser compares folder contents against a team list	No status per team	Submission status per team with filters	Time to produce a missing-submissions list	Established

OP-5	Inconsistent judging	Each judge gets a spreadsheet copy; columns drift	No shared rubric enforcement	One rubric per round, enforced ranges and weights	Number of score-entry errors found during result verification	Established
OP-6	Pending evaluations invisible	Organiser messages judges to ask what is left	No assignment tracking	Assignment matrix with per-judge progress	Time from judging deadline to complete evaluation set	Established
OP-7	Results computed and verified by hand	Copying judge sheets into a master sheet with formulas	Formula mistakes, no audit trail	Deterministic calculation, tie detection, approval step	Time from last evaluation to published results	Established
OP-8	Access control is ad hoc	Drive folders shared with "anyone with link"	Links leak; no revocation tied to roles	Role-based permissions per event, server-enforced	Unauthorised access incidents (target 0)	Established
OP-9	Certificates and feedback are separate manual jobs	Mail merge from a sheet; separate feedback Google Form	Names re-typed; eligibility checked by hand	Certificates generated from platform records; feedback forms in-platform	Time to release certificates after results	Established

OP-10	Organisers will still want Google Sheets for ad-hoc work	Sheets used for counts, calls and sharing with faculty	n/a	One-way Sheets push and CSV export	% of pilot organisers still keeping a parallel manual sheet	Hypothesis

2.2 Participant problems

ID	Problem	Current workflow	Why current tools fall short	Proposed improvement	Expected measurable outcome	Status
PP-1	Rules and schedule scattered across PDFs, posts and chats	Ask in WhatsApp group	PDF goes stale; messages get buried	Single published guidebook plus announcements on the team dashboard	Repeated rule/schedule questions to organisers (pilot observation)	Established
PP-2	Unsure whether submission was received	Email organiser to confirm	Forms/Drive give no team-level status	Submission status and timestamp visible to all team members, with email receipt	Submission-status queries to organisers	Established
PP-3	Last-minute upload failures	Upload to Drive near deadline	No retry, no clarity on what counts	Draft saving, clear required fields, server timestamp, failure messages	% of teams that start a submission and fail to submit	Hypothesis

PP-4	Certificates arrive late or misspelt	Wait for mass email	Names re-typed by organisers	Name-on-certificate field set by participant, downloadable certificates	Certificate correction requests	Established

2.3 Judge problems

ID	Problem	Current workflow	Why current tools fall short	Proposed improvement	Expected measurable outcome	Status
JP-1	Hunting for the right files	Open Drive folders and match by name	Files not linked to rubric or team	Evaluation workspace showing submission beside the rubric	Median time to evaluate one team	Established
JP-2	Unclear criteria	Rubric in a separate PDF	Descriptions not visible while scoring	Criterion descriptions and ranges on the scoring form	Judges asking organisers what a criterion means (observation)	Hypothesis
JP-3	Lose track of which teams are done	Personal notes	No status	Assigned-team list with status	% evaluations completed by the judging deadline	Established

2.4 Helping staff problems

ID	Problem	Current workflow	Why current tools fall short	Proposed improvement	Expected measurable outcome	Status
SP-1	Check-in done against printed lists	Tick names on paper, later typed up	Not live; organisers cannot see who arrived	Mobile check-in by team ID or name, visible to admins in real time	Time to check in one team (observation)	Hypothesis
SP-2	Volunteers given full spreadsheet access	Share the master sheet	Exposes phone numbers and scores	Staff view limited to fields needed for the task	Staff-visible fields audited against matrix (Section 8)	Established
SP-3	Issues reported informally	WhatsApp messages to organisers	Lost in chat; no status	Issue report with category and escalation status	Issues with no recorded resolution at event end	Hypothesis

2.5 Hypotheses to validate with real organisers

Validate these with at least 3 organising teams (target, not a requirement) before Phase 3 starts. Do not treat any as confirmed until evidence exists.

ID	Hypothesis	How to validate	Decision it affects
H-1	Organisers will accept the platform as the source of truth and stop editing the master Google Sheet	Interview; observe pilot	Whether bidirectional sync is ever needed

H-2	A one-way, snapshot-style Sheets push is enough for organiser needs	Show prototype sheet; ask what they would edit	Sheets design
H-3	Judges do not need Google accounts if files are viewed inside the platform	Ask judges; test in pilot	Drive vs native storage
H-4	Most target events use a single round; multi-round events advance teams by organiser selection, not automatic cut-off	Review 5 past event structures	Multi-round priority
H-5	Arithmetic mean of judge scores without normalisation is acceptable to organisers	Interview faculty coordinators	Need for judge-bias normalisation
H-6	Anonymous (blind) judging is not required for most target events	Interview case competition organisers specifically	Anonymous judging priority
H-7	Join-code team formation is understood without help	Usability test with 5 students	Team formation UX
H-8	Staff tasks are mostly check-in and verification; a general task board is unnecessary	Interview past volunteers and coordinators	Staff module scope
H-9	Participants accept that feedback marked "confidential" is not fully anonymous	Ask in usability test	Feedback anonymity design

H-10	A background image plus placed text fields is enough for certificate templates	Collect 5 past certificates and check	Certificate builder scope
H-11	Organisers want judge comments shared with teams after results	Interview	Feedback sharing priority

3. Product goals and non-goals

V1 succeeds if one real event runs end to end on the platform with correct, defensible results; time savings are measured against a baseline captured during the pilot, not promised in advance.

3.1 Goals

No historical baselines exist. Where a target says "baseline", the team records the organiser's current effort (by interview or diary) before the pilot and compares after. Section 12 lists the full pilot metric set; the goals below are the product-level commitments.

ID	Goal	Definition	Measurement method	Target / baseline requirement	Collection
G-1	Faster event configuration	Elapsed active time from "Create event" to event published with guidebook, one round and registration open	Event created_at to published_at timestamps, plus observer timing to exclude idle time	Baseline: establish in pilot. Target: a first-time organiser completes setup in one sitting without developer help	Automatic + observation

G-2	Less manual registration effort	Organiser hours spent correcting participant/team records during registration	Count of admin edits to participant/team records (audit log) + organiser self-report	Baseline: organiser's estimate of hours for their last event. Target: fewer hours than baseline	Automatic + survey
G-3	Faster judging	Median time a judge spends per team evaluation	First open of evaluation to submit, excluding gaps over 15 minutes	Baseline: establish in pilot (no prior data)	Automatic
G-4	No submission mismatches	Submissions associated with the wrong team	Organiser/judge-reported mismatches confirmed in post-event review	Target: 0	Manual review
G-5	Faster results	Time from judging deadline to results published	Round judging_deadline to result published_at	Baseline: organiser estimate from last event. Target: below baseline	Automatic + interview
G-6	Higher submission completion	Registered complete teams that submit before the deadline	Submitted teams / complete teams at registration close	Baseline: establish in pilot	Automatic

G-7	Correct results	Published results needing correction after publication	Count of republications caused by calculation or data errors	Target: 0 caused by calculation errors	Automatic (audit log)

3.2 Non-goals for V1

Non-goal	Reason
Payments and registration fees	Adds compliance and refund handling; brief excludes complex payments
QR badges and QR check-in	Search-based check-in is enough for the pilot scale
Public leaderboards and live score displays	Risk of exposing unapproved scores; not needed to run an event
AI features (auto-judging, summarisation, chatbots)	No validated need; adds cost and trust risk to scoring
Sponsorship CRM and mentor scheduling	Outside core event operations
Bidirectional Google Sheets sync	Conflict resolution is costly and error-prone; see 6-F
Multi-tenant organisation administration, SSO, billing	Single organising team per event is assumed (AS-06)

Complex analytics and custom report builders	Operational dashboard and CSV export cover pilot needs
Advanced notification automation (scheduled reminders, SMS, WhatsApp)	Only essential transactional emails in V1
Automatic judge-bias normalisation	Hard to explain to participants; validate need first (H-5)
Anonymous/blind judging	Not required for most target events (H-6, to validate)
Native mobile apps	Responsive web is sufficient
Offline mode	Venue connectivity assumed intermittent, not absent (AS-05)
Public event discovery marketplace	Events are reached through a shareable link

4. User personas and jobs to be done

Five personas cover the four interfaces; the organiser interface has two personas because student leads and faculty coordinators need different things from it. Personas are illustrative composites, not research findings.

4.1 Persona summary

Attribute	Riya: Student organiser lead (Core Admin)	Dr. Mehta: Faculty coordinator	Arjun: Team leader (Participant)	Ms. Kapoor: Industry	Sahil: Volunteer

		(Core Admin)		judge (Judge)	(Helping Staff)
Role	Convenor of the coding club running a 24-hour hackathon with 3 tracks	Faculty in charge; signs off results and certificates	Third-year student leading a 4-person team	Product manager judging 12 teams in the final round	Second-year student handling registration desk and check-in
Responsibilities	Event setup, rules, registrations, judge coordination, results	Approve results, handle disputes, sign certificates	Register team, submit project, keep team informed	Review submissions, score fairly, give useful comments	Check in teams, verify IDs, report issues, run errands
Goals	Run the event without spreadsheet firefighting; publish results on time	Defensible, error-free results; no complaints	Submit on time; know the rules; get certificate	Finish evaluations within the slot; clear criteria	Clear instructions; quick lookups at the desk
Pain points	Chasing judges; merging sheets; last-minute team changes	Manual totals; no record of who changed what	Unsure if upload worked; rules spread across chats	Opening many links; criteria unclear; no progress view	Paper lists; being asked things he cannot answer

Information needed	Registration counts, incomplete teams, missing submissions, judging progress, sync status	Final rankings, ties, score breakdown, audit history	Deadlines, rules, team status, submission status, results	Assigned teams, submission, rubric with descriptions	Team ID, member names, college, check-in status, task list
Most frequent actions	Search and edit teams; post announcements ; assign judges; monitor dashboards	Review results; approve; release certificates	View dashboard; edit submission; read guidebook	Open next team; score; submit	Search team; mark check-in; update task
Permissions	Full admin for her event	Full admin for the event (could be limited to approval in future)	Own profile, own team, own submission	Assigned teams only, own evaluations	Assigned tasks; limited participant fields; check-in
Success	Results published within the planned window; zero mis-attributed submissions	No calculation errors found after publication	Received submission receipt; downloaded certificate	All assigned teams scored before deadline without help	All teams checked in before opening ceremony

4.2 Scenarios

1. **Hackathon setup (Riya).** Two weeks before the event she creates the event, adds three tracks (FinTech, HealthTech, Open Innovation), sets team size 2–4, writes the guidebook from the default sections, and opens registration. She connects the club's Google account so the faculty can see a live sheet.
2. **Case competition finals (Dr. Mehta).** A two-round case competition: Round 1 is a PDF deck from 60 teams judged by 6 judges online; the top 8 present live in Round 2. He

reviews a tie at position 8, decides with the panel, records the reason, approves the Round 1 shortlist, and later approves final results.

3. **Late teammate change (Arjun).** A teammate drops out the night before. Arjun removes her before registration closes and invites a replacement with the join code. After registration closes, he must ask an organiser to make the change.
4. **Judging slot (Ms. Kapoor).** She has 90 minutes for 12 teams. She opens the queue, reads each deck beside the rubric, scores five criteria, adds a comment visible to the team, and submits. She accidentally submits one evaluation with a wrong score and asks the organiser to reopen it.
5. **Check-in rush (Sahil).** 70 teams arrive in 45 minutes. He searches by team ID on his phone, verifies member names against college IDs, marks each member present, and reports a team whose member is not on the roster.

4.3 Jobs to be done

Persona	When…	I want to…	So that…
Organiser	registration is open	see incomplete teams and fix records in one place	the event starts with a clean roster
Organiser	the deadline passes	see exactly which teams did not submit	I can decide on extensions quickly
Organiser	judging is underway	see which judge still owes which evaluation	I can chase the right person
Faculty coordinator	results are ready	see the score breakdown and any ties before approval	I can defend the results
Participant	I join an event	know rules, deadlines and my team's status from one page	I don't need to ask in chat groups
Participant	I submit	get a timestamped confirmation	I can prove I submitted on time

Judge	I start evaluating	see only my teams with the rubric beside the work	I evaluate quickly and consistently
Staff	a team arrives at my desk	find and check them in within seconds	queues stay short

5. Information architecture and navigation

Everything except the user account and the generic rubric template belongs to exactly one event; users enter an event first and then see a role-specific navigation for that event.

5.1 Object hierarchy (conceptual)

GLOBAL
├── User account (one per email)
├── Generic rubric template (system-provided, read-only, copied into rounds)
└── My Events list (derived from the user's event memberships)

EVENT (one per competition)
├── Event settings (metadata, eligibility, team size, registration fields, time zone)
├── Event memberships (Core Admin, Judge, Helping Staff, Participant)
├── Guidebook (sections; draft and published versions)
├── Announcements
├── Tracks (0..n, optional)
│ └── Problem statements (0..n per track, or event-level if no tracks)
├── Rounds (1..n; ordered)
│ ├── Submission form definition and deadline
│ ├── Rubric (one per round; copied from generic or custom)
│ │ └── Criteria
│ ├── Participating teams (all teams in Round 1; shortlisted teams in later rounds)
│ ├── Judge assignments (judge × team, within the round)
│ │ └── Evaluation → Criterion scores + comments
│ └── Result (computed ranking → approved snapshot → published)
├── Teams
│ ├── Team members (participants)
│ ├── Track / problem statement selection
│ ├── Submissions (one per team per round; versioned)
│ │ └── Submission files and links
│ └── Check-in records
├── Participants (registrations; one per user per event)

├── Staff tasks and issue reports
├── Feedback forms (participant form, judge form) → responses
├── Certificate templates → certificates (per user)
├── Integration connections (Sheets, Drive) → sync jobs
└── Audit log

A participant's certificates across events are shown together under the global "My certificates" page, but each certificate still belongs to one event.

5.2 Entry and event switching

1. After sign-in, every user lands on **My Events**, listing events with their role badge (e.g. "Judge" for Event A, "Participant" for Event B).
2. Selecting an event opens that event's workspace with the navigation for the user's role in that event.
3. A header event switcher returns to My Events. The role is never chosen manually; it is determined by the membership (one role per user per event, Section 8).
4. Participants arriving through a shareable event link (`/e/<event-slug>`) see the public landing page first; after sign-in and registration they land on the team dashboard.

5.3 Moving between event-level and team-level information

- **Admin:** event lists (Teams, Submissions, Evaluations) link to a single **Team detail** page with tabs: Overview, Members, Submissions (per round), Evaluations (per round, admin only), Check-in, History. Breadcrumb: Event › Teams › Team T-0042. Back returns to the list with filters preserved.
- **Judge:** Round › My teams › Team evaluation workspace. Next/previous team controls stay within the judge's own filtered list.
- **Staff:** Tasks › Task detail › linked team card (limited fields). Team lookup is also available from the Check-in screen.
- **Participant:** only one team; the team dashboard is the home page.

5.4 Sitemap by interface

CORE ADMIN (per event)
├── Overview (operational dashboard)
├── Setup
│ ├── Event details & publication
│ ├── Eligibility, team size & registration fields
│ ├── Tracks & problem statements
│ └── Rounds, deadlines & submission forms
├── Guidebook (editor, preview, publish, PDF)
├── Announcements
├── People
│ ├── Participants

```
│   ├── Teams → Team detail
│   ├── Judges
│   ├── Staff
│   └── Admins
├── Submissions (per round)
├── Judging (per round)
│   ├── Rubric
│   ├── Assignments
│   └── Progress & missing evaluations
├── Results (per round: review, ties, approve, publish, shortlist)
├── Staff tasks & issues
├── Feedback (forms, responses, sharing)
├── Certificates (templates, generate, review, release)
└── Settings
    ├── Integrations (Sheets, Drive, sync status)
    ├── Exports (CSV)
    └── Audit history

PARTICIPANT (per event)
├── Event landing page (public)
├── Registration
├── Team dashboard (home: status, deadlines, announcements)
├── Team (members, join code, track/problem selection)
├── Guidebook (view, download PDF)
├── Submissions (per round: form, status, history)
├── Results & feedback received
├── Feedback form
├── Certificates
└── My profile

JUDGE (per event)
├── My rounds
├── My teams (per round; status filter)
├── Evaluation workspace (team info + submission viewer + rubric form)
├── Guidebook (read-only)
└── Judge feedback form

HELPING STAFF (per event)
├── My tasks
├── Task detail
├── Check-in (team search → verify → mark present)
├── Report issue / My issues
└── Guidebook (read-only)
```

Navigation is a presentation of permissions, not the permission model itself: hiding a menu item never replaces a server-side check (Section 8).

6. Detailed functional requirements

Requirements are numbered FR-001 onward in module ranges (A: 001–049, B: 050–089, C: 090–119, D: 120–139, E: 140–169, F: 170–189, G: 190–209, H: 210–229, I: 230–249). Each row is testable on its own; Section 16 expands the most important ones into Given/When/Then tests.

6.0 Conventions that apply to every requirement

To keep tables readable, attributes shared by all requirements are stated once here and not repeated per row.

1. **Permissions.** Every action is checked on the server against the user's membership in that event (Section 8). Unauthorised requests return "You don't have access to this" and are not logged as data changes.
2. **Preconditions.** Unless stated, the user is signed in with a verified email and holds the required role in an event that is not archived.
3. **Concurrent edits.** Editable records carry a version number. Saving a stale version fails with: "This record was changed by {name} at {time}. Reload to see the latest version." No silent overwrite.
4. **Validation.** Errors are shown next to the field and summarised at the top of the form; nothing is saved when any validation fails.
5. **Audit.** Requirements marked Ⓐ write an audit entry (who, when, action, entity, before/after, reason when required).
6. **Time.** All deadlines are evaluated with server time and shown in the event time zone with the zone label (e.g. "23:59 IST").
7. **Text limits.** Unless stated: short text ≤ 120 characters, long text ≤ 5,000 characters, names ≤ 80 characters.

6.A Module A: Core Admin / Organiser

Role for all rows: Core Admin of the event (FR-001, FR-002 and FR-003 apply to any signed-in user).

A.1 Access and event lifecycle

ID	Feature	Problem solved	Behaviour and validation	Failure states	Acceptance criteria	Pri

FR-001	Authentication	One identity across roles and events	Sign-up and sign-in by email + password with email verification, or one-time email sign-in link. Password ≥ 10 characters. Password reset by emailed link valid 30 minutes. Admin rights come from event membership, not a separate login. Google sign-in is P1	Wrong credentials: generic "Email or password is incorrect"; 5 failures in 15 minutes locks sign-in for that account for 15 minutes; expired link: "This link has expired" with resend	Unverified user cannot access any event workspace; reset link works once only	P0
FR-002	My Events dashboard	Users in several events need one starting point	Lists every event the user is a member of, with role badge, status and next deadline. "Create event" button visible to all verified users (pilot may restrict creation via an operator allowlist)	No events: empty state with "Create event" and "Have an event link? Open it"	A user who is Judge in A and Participant in B sees both with correct badges	P0

FR-003	Create event Ⓐ	Replaces scattered setup docs	Required: event name, event type (Hackathon / Case competition / Other), mode (In-person / Online / Hybrid), start and end date-time, time zone, organising institution, contact email. Optional: venue, short description (≤ 300 chars), banner image (P1). Slug auto-generated from name, unique platform-wide, editable until published. Creator becomes event owner and Core Admin. Event starts in Draft with one default round "Round 1"	End before start: "End must be after start"; slug taken: suggests slug-2	Event is saved in Draft, invisible on its public link until published	P0

FR-004	Edit event Ⓐ	Details change before events	All FR-003 fields editable. After publication the slug is locked. Changing start/end dates after registrations exist shows a confirmation and prompts an announcement	Stale version (6.0-3)	Edit is visible on the landing page immediately after save	P0

FR-005	Publication and registration status Ⓐ	Control when participants can see and register	Event status: Draft → Published → Concluded → Archived. Registration status (separate): Not open / Open / Closed, set manually or by optional scheduled open/close times. Publish requires: name, dates, time zone, team size, at least one round with a submission deadline. Unpublished guidebook produces a warning, not a block. Unpublish allowed only with zero registrations. Concluded makes the event read-only for participants except feedback and certificates; Archived makes it read-only for everyone except owner	Publish blocked: list of missing items with links	Public link shows "Registration opens on {date}" when Not open, and the registration button only when Open	P0

FR-006	Co-admin management Ⓐ	Organising is a team job	Owner or any Core Admin invites another Core Admin by email. Only the owner can delete or archive the event or remove the owner role. Last Core Admin cannot leave	Invitee already a Judge, Staff or Participant in this event: blocked (one role per event)	Invited admin gains full admin access after accepting	P0
FR-007	Announcements Ⓐ	Participants ask the same questions repeatedly	Admin posts title + body (≤ 2,000 chars) to audience: All participants / Judges / Staff / Everyone. Shown newest first on dashboards. Optional email copy is P1. Edit and delete allowed; edits show "edited"	Empty title or body: validation error	Participant sees the announcement on the team dashboard; a Judge-only post is not visible to participants	P0
FR-008	Conclude and archive event	Stops edits after the event	Conclude requires published results for the final round or explicit confirmation. Archive available 7 days after conclude (owner only)	n/a	After archive, no write actions succeed for non-owners	P1

A.2 Competition structure

ID	Feature	Problem solved	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-010	Tracks Ⓐ	Events with parallel categories	0–20 tracks per event; name (unique within event, ≤ 60 chars), description, display order. Setting "Track selection required" on/off. Deleting a track that teams selected is blocked until teams are moved	Duplicate name; delete blocked with team count	With zero tracks, participants never see a track selector	P0
FR-011	Problem statements Ⓐ	Teams need clear briefs; case competitions reveal cases at a set time	Title, description (rich text), optional attachment (PDF, ≤ 20 MB), optional track. Optional "visible from" date-time (P1) for timed release. Optional cap on teams per statement (P1). Deleting a selected statement is blocked until teams are moved	Attachment too large or wrong type	A statement with a future "visible from" is hidden from participants and judges until that time	P0 (timed release, cap: P1)

FR-012	Rounds and deadlines Ⓐ	Single and multi-stage events	Every event has ≥ 1 round. Per round: name, order, submission opens at, submission deadline, grace period (0–60 min, default 0), late policy (Reject [default] / Accept and flag), judging deadline, ranking scope (Overall / Per track), required evaluations per team (1–5, default 2). Constraints: opens < deadline ≤ judging deadline. Round 2+ includes only shortlisted teams (FR-044). Changing a deadline after submissions exist requires a reason and prompts an announcement	Invalid ordering; reason missing	Single-round event works without the user ever seeing the word "round" more than once in setup	P0 (multi-round: P1)
FR-013	Eligibility Ⓐ	Organisers need participants to confirm eligibility	Up to 5 declaration statements participants must tick at registration (e.g. "I am a currently enrolled undergraduate student"). Optional allowed email domains list (P1), enforced on registration.	Domain not allowed: "This event is open only to emails ending in {domains}"	Registration cannot be submitted with an unticked declaration	P0 (domain rule: P1)

			Detailed eligibility text lives in the guidebook			
FR-014	Team size Ⓐ	Enforce competition rules automatically	Min 1–10, max 1–10, min ≤ max. If max = 1, each registrant automatically forms a solo team. Reducing max below the largest existing team is blocked, listing those teams	Invalid range; blocked change	A team at max size cannot accept another member (FR-055)	P0
FR-015	Registration fields Ⓐ	Avoid separate Google Forms	Fixed fields: full name, email (from account, read-only), phone, institution, course/programme, year of study, name on certificate (defaults to full name). Up to 10 custom fields: short text, long text, number, single select (≤ 20 options), URL, checkbox; each with label, help text, required flag. After registrations exist: new fields must be optional; fields with data can be hidden but not deleted	Duplicate label; select without options	Custom fields appear in the registration form, participant detail, CSV export and Sheets push	P0

FR-044	Shortlist for next round Ⓐ	Case competitions advance top teams	From an approved result of round N, admin selects teams to advance (default selection: top K, editable). Advancing teams gain access to round N+1 submission form. Non-advancing teams see "Not shortlisted" when the shortlist is published	No approved result: blocked	Only shortlisted teams appear in round N+1 assignments and submissions	P1

A.3 People and records

ID	Feature	Problem solved	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-016	Participant management Ⓐ	One reliable roster	List of participants with status: Registered (no team) / In team / Withdrawn / Removed. Detail shows all fields, team, submissions, check-in, certificates. Admin can remove a participant with a reason: they leave their team, lose event access (Section 8.4), and are excluded from certificates	Removing a team leader requires choosing a new leader first if other members remain	Removed participant cannot open the event workspace; record remains visible to admins as Removed	P0

FR-017	Team management Ⓐ	See and fix team state in one place	Team list: Team ID, name, leader, member count, completeness (meets min size), track, problem statement, submission status per round, check-in status. Admin can rename, change track/statement, and disqualify with reason (excluded from rankings and winner certificates; can be reversed)	Team name already used: blocked	Disqualified team shows a badge everywhere admins see it and is excluded from result calculation	P0
FR-018	Team membership by admin Ⓐ	Fix late changes after participant self-service locks	Admin can create a team, add any registered participant without a team, remove a member, and transfer leadership at any time. Max size is never exceeded; below-min teams are allowed but flagged "Incomplete"	Participant already in another team: blocked with link to that team	All changes appear in team History and the audit log	P0

FR-019	Participant record correction Ⓐ	Typos in names and colleges	Admin edits any registration field except login email. Edits to "name on certificate" after certificates are generated mark the certificate "Needs regeneration" (FR-236)	Email change requested: shows "Login email can't be changed in V1" (see open question OQ-07)	Before/after values recorded in audit	P0
FR-020	Search, filter, sort	Finding a team quickly at the desk or during disputes	Search across name, email, phone, team name, Team ID (partial, case-insensitive). Filters: team completeness, track, problem statement, submission status (per round), check-in, participant status. Sort on any column. Filters persist in the URL	No results: "No matches" with "Clear filters"	Search "t-004" finds team T-0042; filters survive a page reload	P0

A.4 Guidebook

ID	Feature	Problem solved	Behaviour and validation	Failure states	Acceptance criteria	Pri

FR-021	Guidebook editor Ⓐ	Rules scattered across PDFs and chats	Section-based editor pre-filled with default sections: Overview, Eligibility, Schedule, Registration, Tracks & problem statements, Submission requirements, Judging criteria, Prizes, FAQs, Code of conduct, Contacts. Sections can be added, renamed, hidden, reordered. Rich text: headings, bold, italic, lists, links, simple tables; images P1. Draft autosaves every 10 seconds of inactivity after a change. "Publish" creates a numbered version with optional change note; participants, judges and staff see only the latest published version with "Updated {date}"	Autosave failure: banner "Not saved – retrying" and changes kept in the browser tab; leaving the page warns	Edits to the draft are invisible to participants until Publish	P0

FR-022	Guidebook PDF export	Organisers still circulate PDFs	PDF generated from the published version on each publish: event name, version, publish date, page numbers, all visible sections in order. Admin can preview the draft as PDF watermarked "DRAFT". Participants download the published PDF	Generation fails: admin sees error with retry; participants still see the web guidebook and "PDF temporarily unavailable"	Downloaded PDF content matches the published web version, including version number	P0

A.5 Judging setup and monitoring

Scoring rules are specified in Module E; these rows cover the admin screens.

ID	Feature	Problem solved	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-023	Generic rubric selection Ⓐ	Organisers need a sensible default quickly	Admin picks "Hackathon default" or "Case competition default" (FR-141) for a round. The template is copied into the round; edits never change the template	n/a	Selecting a template creates a valid, weight-100 rubric in one action	P0

FR-024	Custom rubric builder Ⓐ	Events have their own criteria	Add, edit, reorder, delete criteria within limits of FR-142. Live weight total shown ("Total: 95% – must equal 100%"). Rubric states: Draft → Active → Locked (FR-153)	See FR-143	Judges cannot open a round whose rubric is not Active	P0
FR-025	Rubric weight validation	Prevent wrong totals	Activation blocked unless weights total exactly 100 and every criterion passes FR-143	"Weights total 95%. Adjust to 100% to activate."	Rubric with weights 30/30/30 cannot be activated	P0
FR-026	Judge management Ⓐ	Judges need accounts without admin rights	Invite by email with name and organisation. States: Invited / Active / Deactivated. Deactivating keeps submitted evaluations and removes access. Resend invite	Email belongs to a participant, staff or admin of this event: blocked (one role per event)	Invited judge can only see the judge interface after accepting	P0

FR-027	Judge assignment Ⓐ	Judges must see only their teams	Per round: assign judges to teams individually, or bulk (selected teams × selected judges; all teams in a track to a judge). Auto-balance (P1): distribute so each team gets the round's required number of judges with even load. Shows per-team assigned count vs required. Conflict flag from judge or admin highlights the assignment for reassignment (FR-097). Unassigning a judge who already submitted requires confirmation; the evaluation is retained but excluded (FR-155)	Assigning a judge to a disqualified team: warning; duplicate assignment: ignored	Judge sees exactly the teams assigned for that round	P0 (auto-balance: P1)

FR-028	Submission monitoring Ⓐ	Know who has not submitted	Per round: counts and list by status (Not started / Draft / Submitted / Late / Missing after deadline). View any version, open files and links. Extend for one team: set a team-specific deadline and reason, which reopens editing for that team only. Bulk ZIP download P1	File unavailable: shown per file with error	Filtering "Missing" after deadline lists exactly teams with no submitted version	P0
FR-029	Judging status control Ⓐ	Stop judging on draft submissions	Per round: judging Not open / Open / Closed. Judging can open only when the rubric is Active and the round's submission deadline (including grace) has passed. Closing judging blocks new evaluation submissions	Open blocked: reason shown	Judges cannot score before submissions lock	P0
FR-030	Judging progress	Chasing judges blindly	Per judge: submitted / assigned; per team: submitted evaluations / required. Highlights judges with zero progress	n/a	Numbers match the evaluation records exactly	P0

			1 hour before judging deadline			
FR-031	Missing evaluation detection Ⓐ	Results cannot be finalised with gaps	Lists every assignment without a submitted evaluation and every team below required evaluations. Actions: reassign to another judge, exclude assignment with reason, remind judge (email P1)	n/a	After judging closes, the list is empty or every item is excluded with a reason before approval is allowed	P0

A.6 Results, feedback and certificates

ID	Feature	Problem solved	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-032	Result review	Manual verification of totals	Computed ranking per round (FR-150) with per-team breakdown per judge and criterion. Flags: Tie, Incomplete, Disqualified, Excluded evaluations	n/a	Breakdown values reproduce the team score by hand using the formula in 6-E	P0

FR-033	Tie resolution Ⓐ	Ties at award positions need a decision	Ties at positions within the configured number of award positions (default 3 per ranking scope) must be resolved before approval: either "Share position" or "Order manually" with a reason ≥ 10 characters	Approve blocked until resolved	Resolution and reason appear in the approved snapshot	P0
FR-034	Result approval Ⓐ	Faculty sign-off	Preconditions: judging Closed; no unexcluded missing evaluations; award-position ties resolved. Approval creates an immutable, versioned snapshot (ranks, scores, breakdown, approver, time). Any later evaluation change or disqualification invalidates the approval and returns the result to Review	Blocked with checklist of unmet preconditions	Snapshot is unchanged by later edits; invalidation is shown with cause	P0

FR-035	Result publication Ⓐ	Announce results reliably	Publish the latest approved snapshot. Visibility options: winners list (on), each team sees its own rank (off), scores shown (off). For non-final rounds, publish the shortlist instead (P1). If a published result is invalidated, participants keep seeing the last published version until a new version is approved and republished, labelled "Updated {date}". Unpublish requires a reason	Publish without approval: not possible	Participants see results only after Publish; admins see an audit entry	P0
FR-036	Feedback management	See Module H (FR-210 to FR-219)				P0
FR-037	Certificate templates	See Module I (FR-230 to FR-239)				P0
FR-038	Certificate generation and release	See Module I				P0

A.7 Data, integrations, operations and audit

ID	Feature	Problem solved	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-039	CSV export Ⓐ	Organisers need data outside the platform	Exports: participants, teams with members, submissions (latest version, links, timestamps), evaluations per criterion, results (approved snapshot), check-in, feedback responses. UTF-8 with BOM, one header row, ISO 8601 timestamps plus event-zone column. Cells starting with = + - @ are prefixed with an apostrophe to prevent formula injection. Exports reflect current filters	Export over 10,000 rows runs in background and notifies when ready	Exported participant count equals list count for the same filters	P0
FR-040	Google Sheets settings	See Module F				P1
FR-041	Google Drive settings	See Module G				P1

FR-042	Sync status and errors	One place to see integration health	Settings › Integrations shows each connection: status (Connected / Needs reauthorisation / Disconnected), last successful sync time, last attempt result, pending jobs, error list with plain-language message and "Retry". Dashboard card shows a red state if the last attempt failed	n/a	A failed sync is never displayed as successful (FR-180)	P1
FR-043	Operational dashboard	One glance at event health	Cards: participants and teams (complete / incomplete / no team), submissions per round by status, judging progress, check-in progress, open staff issues, integration health, next three deadlines. Each card links to the filtered list. Data refreshed on load and by a refresh button; auto-refresh every 60 s is P1	A card failing to load shows its own error; others still load	Every number on the dashboard equals the count on the linked list	P0
FR-045	Staff management and task	Volunteers need direction without	Invite staff by email. Create tasks and assign to one or more staff (Module D). View issue	Email already holds another role in this	Staff see only tasks assigned to them	P0

	assignment Ⓐ	admin access	reports and resolve them	event: blocked		
FR-046	Audit history	Defend decisions and trace mistakes	Read-only list of all Ⓐ actions, filterable by entity type, user, and date range; exportable to CSV. Entries cannot be edited or deleted by any role. Retained for the life of the event record	n/a	Changing a judge assignment produces an entry with actor, time, round, team, judge, old and new state	P0

6.B Module B: Participant / Team interface

Role for all rows: Participant of the event, acting on their own record and team only. Participant self-service for team changes ends when registration closes; after that, only Core Admins change membership (FR-018).

B.1 Access, registration and teams

ID	Feature	Problem solved	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-050	Event access by link	Participants need one entry point	Public landing page at the event link: name, dates and time zone, mode, venue, short description, tracks, link to published guidebook, registration status and the primary action (Register / Go to my team /	Draft or archived event: "Event not found"	Landing page loads without an account; registration button hidden when registration is not Open	P0

			Registration closed). No sign-in needed to view			
FR-051	Participant registration Ⓐ	Replace Google Forms	Requires sign-in. Form: fixed fields (FR-015), custom fields, eligibility declarations. One registration per user per event. On success: status Registered, confirmation email, redirect to "Create or join a team". A participant who withdrew may register again while registration is Open; a participant removed by an admin cannot	Registration not open: "Registration is closed"; already registered: redirected to team dashboard; missing required field	Submitting twice (double click) creates one registration	P0

FR-052	Team creation Ⓐ	Teams form without organiser effort	Registered participant without a team enters a team name (3–40 chars, unique within event, case-insensitive). Creator becomes leader. System assigns an immutable Team ID ("T-" + 4-digit sequence per event, never reused) and an 8-character join code (uppercase letters and digits, excluding 0, O, 1, I). Leader can regenerate the code, which invalidates the old one. Allowed only while registration is Open	Name taken: "A team with this name already exists"; already in a team: blocked	New team appears in admin team list immediately with leader as only member	P0

FR-053	Join existing team Ⓐ	Members join without organiser help	Enter join code → confirm screen showing team name, leader name, current size → Join. Join is atomic: the size check and insert happen together so two simultaneous joins cannot exceed max	Full: "Team {name} is full ({n}/{max}). Ask the leader or create your own team." Already in a team: "You're already in {team}. Leave it before joining another." Invalid code: "Code not recognised." 10 failed attempts in 10 minutes: "Too many attempts – try again in 10 minutes." Registration closed: blocked	Two users joining the last slot at the same time: exactly one succeeds	P0

FR-054	Team membership management Ⓐ	Teams change before the event	While registration is Open: leader can remove members and transfer leadership; members can leave. A leader cannot leave while other members remain without transferring leadership first. If the last member leaves, the team is dissolved (Team ID retired). A member who leaves loses access to the team's submissions; submissions stay with the team. After registration closes all buttons are replaced by "Contact the organisers to change your team"	Remove self as leader: blocked with transfer prompt	Team below min size shows "Incomplete – needs {k} more member(s)" to all members	P0
FR-055	One team per event	Prevent duplicate membership	Unique constraint: a participant belongs to at most one team per event, enforced in the	See FR-053	Direct API attempt to add a member already in a	P0

			database, not only the UI		team is rejected	
FR-056	Participant profile Ⓐ	Keep own details correct	View and edit own registration fields while registration is Open. After close: phone and "name on certificate" stay editable until certificates are generated for this user; other fields are read-only with "Contact organisers to correct"	Invalid phone format (7–15 digits, optional +)	Edits reflect in admin views and next Sheets push	P0
FR-057	Team profile and Team ID	Teams and organisers need a shared reference	Team page shows Team ID (copyable), name (leader can edit until registration close), members (name, institution; teammates also see each other's email), leader badge, track and problem statement, completeness, join code (members only, until registration close)	n/a	Team ID shown on dashboard, submission receipt, check-in and certificates	P0

FR-058	Withdraw from event Ⓐ	Participants drop out	While registration is Open, a participant can withdraw; they leave their team (leader rules as FR-054). After close, withdrawal is by admin only	n/a	Withdrawn participant is excluded from certificates and team counts	P1

B.2 Information and selection

ID	Feature	Problem solved	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-059	Team dashboard	One place for status	Shows: team status (complete/incomplete), track/statement, next deadline with countdown in event time zone, per-round submission status, latest 5 announcements, results when published, certificate availability	Section fails to load: that section shows a retry, others render	A team with a draft but no submission sees "Draft – not submitted" in red	P0
FR-060	Guidebook view and download	Rules in one place	Read the latest published guidebook with section navigation and "Updated {date}"; download PDF (FR-022)	Guidebook not yet published: "Rules will be published soon"	Participant never sees draft content	P0

FR-061	Track selection Ⓐ	Teams pick their category	Leader selects a track; members see it. Required if "Track selection required" is on, before the team can submit. Changeable until the Round 1 submission deadline	No track chosen but required: submit blocked with "Choose a track first"	Changing track after judge assignment shows a warning to admins (assignment may need review)	P0
FR-062	Problem statement selection Ⓐ	Teams pick a brief	Leader selects among statements visible to them (filtered by chosen track). Same lock rule as FR-061. Cap per statement is P1: full statements show "Full" and cannot be chosen	Statement full (P1)	Statement with future "visible from" is not selectable	P0
FR-063	Event updates	Avoid chat-group dependency	Announcements feed (FR-007) on dashboard and a dedicated page; unread marker per user	n/a	New announcement shows as unread until opened	P0

B.3 Submissions

The admin defines each round's submission form in round setup. Field types: project title (fixed, required), summary (fixed, long text ≤ 1,500 chars, required), file, repository URL, demo video URL, live URL, other URL, short text, long text. Each configurable field has a label, help text, required flag. Maximum 10 configurable fields.

Default forms (editable): **Hackathon** = title, summary, presentation file (PDF/PPT/PPTX, required), repository URL (required), demo video URL (optional), live URL (optional). **Case competition** = title, summary, deck file (PDF, required). **Minimum viable submission** is title + summary + all fields the organiser marked required.

ID	Feature	Problem solved	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-064	Submission requirements view	Teams miss requirements	Per round: list of fields with required markers, accepted types, size limits, deadline, grace period and late policy stated in plain words	n/a	Requirements shown match the round configuration exactly	P0
FR-065	Structured submission form Ⓐ	Files and links scattered	Any team member can edit the draft and submit. Draft saves on each field change. "Submit" validates all required fields and records submitted_by and server timestamp. Only teams meeting min size and with required track/statement can submit; admins can grant an override per team with reason	Incomplete: "Complete the required fields: {list}" and draft kept; team incomplete: "Your team needs at least {min} members to submit – contact organisers"	Submitted version is read-only and is the one judges see	P0

FR-066	File upload	Decks and reports	Allowed per field: PDF, PPT, PPTX (organiser may also allow PNG, JPG, ZIP). Max size per file: default 25 MB, configurable 1–50 MB. One file per field in V1 (multiple files: P1). Type validated by extension and file signature. Progress bar; on failure the field shows the error and "Retry"; a draft is never marked complete with a failed file. Replacing a file in a draft keeps the old file in the previous submitted version	Too large: "File is {x} MB; the limit is {y} MB"; wrong type; network failure: "Upload failed – your other answers are saved. Retry."	A PPTX renamed to .pdf is rejected	P0

FR-067	Repository URL	Judges need code	Must be https and match github.com, gitlab.com or bitbucket.org with owner and repo path. Help text: "Make sure the repository is public or shared with the organisers." Reachability check that warns (does not block) if the URL returns an error: P1	Invalid format	"github.com/user" without repo is rejected	P0
FR-068	Demo video and live URL	Judges need demos	Valid http(s) URL, ≤ 2,000 chars. Help text reminds teams to set videos public or unlisted. No content check in V1	Invalid URL	Links open in a new tab for judges	P0
FR-069	Additional fields	Event-specific needs	Organiser-defined short text, long text or URL fields within the 10-field limit	Validation per type	Shown to judges in configured order	P0

FR-070	Submission timestamps	Proof of timeliness	Server records first_submitted_at and submitted_at for every version. Team sees "Submitted {date time zone} by {name}" and a version history. Submission receipt email to all members	Email failure does not affect submission success; receipt visible on screen	Timestamp displayed equals server record	P0
FR-071	Deadline enforcement	Fairness	Submit accepted if server time ≤ deadline + grace, or ≤ team extension. After that: Reject policy → form read-only with "The deadline passed at {time}. Contact organisers if you believe this is an error." Accept-and-flag policy → accepted, status "Late by {n} min". Drafts are never auto-submitted (open question OQ-04). Submissions in progress when the deadline passes are	Upload started before deadline but completed after: treated by completion time; shown clearly in help text	Submit request at deadline + grace + 1 s under Reject policy fails	P0

			judged by server receipt time of the final request			
FR-072	Edit before deadline Ⓐ	Teams improve work until the deadline	Editing a submitted submission creates a new draft; the last submitted version remains official until "Resubmit". Banner: "You have unsubmitted changes. Your last submitted version (v{n}) will be judged unless you resubmit." Concurrent edits by two members use version checks (6.0-3)	Stale save: reload prompt	Discarding the draft restores the last submitted version	P0
FR-073	Submission status	Teams and organisers know where they stand	Status: Not started / Draft / Submitted / Submitted late / Locked (after deadline). Version list with timestamps and submitter	n/a	After deadline, status changes to Locked for teams with a submitted version and "Not submitted" otherwise	P0

FR-074	Results and shared feedback	Teams want outcomes	After publication: winners list and, if enabled, own rank. Judge comments marked shareable appear after the admin releases them (FR-216), without judge names	Not published: "Results will appear here once published"	Team never sees another team's comments or any score unless enabled	P0
FR-075	Participant feedback	See Module H				P0
FR-076	Certificate access	See Module I				P0

B.4 Required behaviour in specific situations

Situation	System behaviour
Joining a full team	Blocked with message in FR-053; no partial membership created
Joining a second team in the same event	Blocked by unique constraint (FR-055); message names the current team
A team member leaves	Membership ends; team completeness recalculated; if below min, all members see "Incomplete"; submissions remain with the team; the member's certificate eligibility for this team ends

Incomplete submission	Draft retained; Submit disabled until required fields complete; status "Draft – not submitted" on dashboard and admin list
Deadline passed	Form locked; behaviour per late policy (FR-071); extension only by admin (FR-028)
File upload fails	Field shows error and retry; other fields remain saved; Submit blocked until a valid file is present for required file fields
Editing a locked submission	All inputs read-only; message "Submissions closed at {time}"; API rejects writes
Certificate not yet released	Certificates page shows "Certificates will be available after the organisers release them"; no download links exist

6.C Module C: Judge interface

Role for all rows: Judge of the event, acting only on their own assignments. Judges never edit event configuration, rubric weights or participant records, never change another judge's scores, and never publish results; these are enforced server-side (Section 8).

Anonymous judging (Recommendation). Not in V1. College hackathons typically judge with team names visible, and live pitches reveal identity anyway. Case competitions sometimes require blind review; this is hypothesis H-6. If confirmed, the P2 option is "Hide member names and institutions from judges" per round.

Teams without a submission. When a round's deadline passes, teams with no submitted version get status "Did not submit". Their assignments become "Not required", they are excluded from that round's ranking, and judges see them greyed out. An admin who grants a late extension restores the assignments.

ID	Feature	Problem solved	Behaviour and validation	Failure states	Acceptance criteria	Pri

FR-090	Judge authentication	Judges need quick, separate access	Invite email → create password or use email sign-in link (FR-001). Accepting the invite activates the Judge membership	Expired invite (7 days): "Invite expired – ask the organiser to resend"	Judge lands on My rounds for that event	P0
FR-091	Assigned rounds	Judges may judge several rounds	Lists rounds with assignments: round name, judging status (Not open / Open / Closed), judging deadline, progress "{submitted} of {required}"	Judging not open: rounds visible, scoring disabled with "Judging opens after submissions close"	Judge with no assignments sees "You have no teams assigned yet"	P0
FR-092	Assigned team list and status	Judges lose track	Per round: Team ID, team name, track, problem statement, submission status (Submitted / Late / Did not submit), evaluation status (Not started / Draft / Submitted / Reopened / Not required / Conflict). Search by team name or ID; filter by status and track; default sort: Not started first	Empty filter result: "No teams match"	List contains only teams assigned to this judge in this round	P0

FR-093	Team profile in workspace	Context for scoring	Shows Team ID, name, members (name, institution), track, problem statement (title, expandable description), project title and summary	Missing optional fields hidden, not shown blank	Judge sees no participant phone, email or custom fields	P0
FR-094	Submission viewer	Avoid hunting for files	Files from the evaluated (last submitted) version with version number and timestamp. PDFs open in an in-page viewer beside the rubric; PPT/PPTX download (in-page preview P1). Links (repository, video, live) open in a new tab. Late flag shown. Files are served through short-lived authorised links (FR-195)	File cannot load: "This file could not be loaded. Retry or report to organisers" with "Report missing file" that notifies admins	Judge can score while a file is reported missing; admin sees the report	P0

FR-095	Rubric scoring form	Consistent scoring	Criteria in configured order with name, description, weight (%) and range. Input accepts integers within the criterion range only (half points: P2). Shows the judge's live weighted score (0–100, 2 decimals) using FR-146. Comment fields: "Feedback for the team" (optional or required per rubric; may be shared with the team, FR-216) and "Private note to organisers" (optional, never shared)	Out-of-range or non-integer value: field error "Enter a whole number from {min} to {max}"	Entering 11 on a 1–10 criterion is rejected client- and server-side	P0
FR-096	Draft evaluations	Judges get interrupted	Autosave on each change; status Draft; resumable from any device. Drafts are never counted in results	Autosave fails: "Not saved – retrying"; unsaved values kept in the tab; navigation warns	Closing and reopening the browser restores the draft	P0

FR-097	Conflict of interest Ⓐ	Fairness	Before submitting, a judge can mark "I have a conflict of interest" with a reason (≥ 10 chars). Scoring is disabled, the assignment shows Conflict, and admins see it on the judging progress screen to reassign. Admins can also mark a conflict. Admin may override a judge-declared conflict only with a reason	n/a	A conflicted assignment never contributes to results	P0
FR-098	Evaluation submission Ⓐ	Clear completion	Submit validates: every criterion scored, values valid, required comment present, judging Open, assignment active. Confirmation dialog lists all scores and the weighted score: "Submit evaluation for {team}? You can't edit it after submitting." On success: status Submitted, server timestamp,	Judging closed: "Judging for this round has closed"; assignment removed meanwhile: "You are no longer assigned to this team"; validation errors listed	Submitted evaluation is read-only for the judge	P0

			confirmation screen with "Next team"			
FR-099	Duplicate evaluation prevention	Double clicks and two tabs	One evaluation per assignment (database constraint). Submit is idempotent: a repeated request with the same submission token returns the existing result	Second tab submitting different values after the first: "This evaluation was already submitted at {time}"	Double-clicking Submit produces one evaluation and one audit entry	P0
FR-100	Evaluation correction Ⓐ	Accidental submissions happen	Judge clicks "Request correction" with reason; admin approves or declines. Approval (or admin-initiated reopen with reason) sets status Reopened, keeps the previous version in history, and lets the judge edit and resubmit. Admins never type scores on a judge's behalf. Reopening after result approval invalidates the approval (FR-034). Reopen is allowed after	Declined: judge sees the admin's reason	Results recalculate from the resubmitted version; both versions visible to admins	P0

			judging closes for that one evaluation only			
FR-101	Status updates	Judges see progress	After submit, list and round progress update on return; no real-time push required	n/a	Progress counter equals count of Submitted evaluations	P0
FR-102	Access restrictions	Protect other teams' data and scores	Judge can read only: assigned teams' profiles and submissions in rounds where they are assigned, their own evaluations, the active rubric, and the published guidebook. Cannot see other judges' scores or comments, aggregate scores, or results before publication. Requests outside scope return "not found" (no confirmation the record exists)	n/a	Requesting another team's submission by ID returns not found and is logged as a denied access	P0
FR-103	Judge feedback form	See Module H				P0

6.D Module D: Helping Staff interface

Role for all rows: Helping Staff of the event. The interface is a task list plus a check-in tool, designed for phones. It is deliberately not a task-management system: tasks have one status, no subtasks, no comments thread.

Staff-visible participant data. Full name, Team ID, team name, institution, year of study, track, check-in and ID-verification status. Email is never shown. Phone is shown only if the admin enables "Staff can see phone numbers" for the event (P1, default off). Custom registration fields are hidden unless the admin marks a field "Visible to staff" (P1). Staff never see submissions, scores, judge comments, rubrics, results before publication, feedback responses, integration settings or audit history.

Staff can modify only: status of tasks assigned to them, check-in and ID-verification marks (with a note), and issue reports they created.

ID	Feature	Problem solved	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-120	Staff authentication	Volunteers need limited access fast	Invite email → account → Staff membership (FR-090 rules)	Expired invite	Staff lands on My tasks	P0
FR-121	Task creation (admin side) Ⓐ	Direct volunteers	Admin creates a task: title (required), description, type (Check-in desk / Verification / General), assigned staff (1+), optional due time, optional location, optional linked teams (0–50)	No staff assigned: validation error	Task appears only for assigned staff	P0

FR-122	My tasks	Know what to do	List of tasks assigned to me with status, due time, location, linked team count; filter by status; overdue tasks highlighted	No tasks: "No tasks assigned yet"	Staff never see unassigned tasks	P0
FR-123	Task status Ⓐ	Organisers see what is done	Status: To do → In progress → Done, or Blocked (requires a note). Any assigned staff can change it; last change shows who and when. Blocked tasks appear on admin dashboard	Missing note for Blocked	Admin sees updated status on next load	P0
FR-124	Team lookup	Fast desk operations	Search by Team ID, team name or participant name (partial). Results show the limited team card; staff can look up any team in the event (needed at an open check-in desk)	No match: "No team found – check the Team ID or report an issue"	Lookup returns no email addresses	P0

FR-125	Participant verification Ⓐ	Confirm the right people are present	Per member: mark "ID verified" with optional note (≤ 200 chars). Staff cannot edit names or team membership; a mismatch is reported as an issue	n/a	Verification shows who verified and when	P0
FR-126	Team check-in Ⓐ	Live attendance	Mark each member present; team status derived: Not checked in / Partially checked in / Checked in. Undo allowed for the staff member who marked it or any admin, logged. Admin dashboard shows counts	Duplicate tap: idempotent; network failure: "Not saved – retry" and the toggle reverts	Two staff checking in the same member results in one check-in record	P0
FR-127	Queue or waiting status	Crowd management	Not in V1; recommended P2 if pilot shows desk queues are a problem			P2

FR-128	Issue reporting Ⓐ	Problems lost in chat	Fields: category (Registration mismatch / Technical / Facilities / Conduct / Other), description (required), optional linked team, severity (Normal / Urgent). Goes to the admin issue queue. Status: Open → Acknowledged → Resolved (admin sets; resolution note required). Staff see status of issues they reported	Empty description	Reported issue appears in admin queue with reporter and time	P0
FR-129	Escalation	Urgent problems need fast attention	"Escalate" on an Open issue sets severity Urgent; Urgent issues are pinned at the top of the admin dashboard with a count. Email or push alert to admins: P1	n/a	Escalated issue is visibly distinct on admin dashboard	P0 (alerts : P1)
FR-130	Task completion confirmation	Avoid ambiguity	Marking Done shows "Task marked complete" with optional completion note; admin sees	n/a	Done tasks move to a Completed filter	P0

completion time
and person

6.E Module E: Judging and scoring engine

The engine is deliberately simple: every criterion is required, each score is divided by its criterion maximum, weights total exactly 100, and a team's score is the plain mean of its counted judge scores. Nothing is averaged, normalised or filled in silently.

E.1 Formula

For a rubric with criteria i = 1…k, each with integer weight w_i (percent), integer range [min_i, max_i], and a judge's integer score s_i:

$$n_i = \frac{s_i}{\max_i} \qquad \text{(criterion score as a fraction of its maximum)}$$
$$J = \sum_{i=1}^{k} w_i \cdot n_i \qquad \text{where } \sum_{i=1}^{k} w_i = 100 \text{, so } J \in [0, 100]$$
$$T = \frac{1}{m} \sum_{j=1}^{m} J_j \qquad \text{over the } m \ge 1 \text{ counted evaluations of the team in that round}$$

Normalisation decision (Recommendation). Scores are scaled by the criterion maximum (s/max), not min–max scaled. Reason: a 7 on a 1–10 criterion reads as 70%, matching how organisers check scores in spreadsheets. The brief's example (divide by the sum of weights) gives the same result as above because weights always total 100; normalising first is what makes criteria with different ranges comparable. To keep s/max meaningful, min_i must be 0 or 1.

Precision. Calculations use exact decimal arithmetic with no intermediate rounding. Stored and compared values are rounded half-up to 4 decimal places; displayed values to 2 decimal places.

Worked example (use as a test fixture). Criteria: A weight 30, range 1–10; B weight 40, range 1–10; C weight 30, range 0–5.

Evaluation	s(A)	s(B)	s(C)	J calculation	J
Judge 1	8	7	4	30×0.8 + 40×0.7 + 30×0.8	76.0000
Judge 2	6	9	3	30×0.6 + 40×0.9 + 30×0.6	72.0000

Team score T	(76 + 72) / 2	74.0000

E.2 Requirements

ID	Feature	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-140	Rubric structure	One rubric per round. Rubric: name, state (Draft / Active / Locked), criteria (ordered), comment settings (team feedback field: Off / Optional / Required). Criterion: name (≤ 60 chars, unique in rubric), description (≤ 500 chars), weight (integer 1–100), min (0 or 1), max (integer 2–100, > min)	n/a	A round cannot open judging without an Active rubric	P0
FR-141	Default generic rubrics	**Hackathon default** (range 1–10 each): Problem understanding & relevance 20; Innovation & originality 20; Technical execution 25; Impact & feasibility 20; Presentation & demo 15. **Case competition default** (1–10 each): Problem analysis 25; Quality of recommendations 25; Feasibility & implementation 20; Quantitative/financial rigour 15; Presentation & communication 15. Each criterion ships with a one-sentence description	n/a	Each template activates without edits	P0

FR-142	Custom rubric limits	1–12 criteria. Criteria may have different ranges (e.g. 0–5 and 1–10)	13th criterion: "A rubric can have at most 12 criteria"	Limits enforced on save	P0
FR-143	Rubric validation	Activation requires: ≥ 1 criterion; weights total exactly 100; each weight 1–100; min ∈ {0, 1}; max > min; unique names. Drafts may be saved invalid	Specific message per rule (e.g. "Weights total 95%; they must total 100%")	Each rule has a negative test (Section 16)	P0
FR-144	Required and optional criteria	All criteria are required in V1. Optional criteria are P2 because they require a defined rule for re-weighting, which the brief warns against introducing silently	n/a	Evaluation cannot be submitted with any criterion blank	P0
FR-145	Score validation	Each score must be an integer within $[min_i, max_i]$. Validated in the browser and again on the server; invalid submissions are rejected whole, nothing saved	"Enter a whole number from {min} to {max}"	API request with s = 10.5 or 11 on a 1–10 criterion returns a validation error and changes nothing	P0
FR-146	Judge weighted score	J computed by E.1 at submission and stored with the evaluation, alongside raw scores	n/a	Worked example yields 76.0000 and 72.0000	P0

FR-147	Team round score	T = mean of J over counted evaluations. Counted = Submitted, assignment active, not Conflict, not Excluded, judging round matches. Drafts and Reopened evaluations are not counted until resubmitted	n/a	Worked example yields 74.0000; adding a Draft from a third judge leaves T unchanged	P0
FR-148	Missing-score handling	A partly scored evaluation can only be a Draft. No imputation, no partial averaging	Submit blocked listing unscored criteria	No code path produces J from fewer than all criteria	P0
FR-149	Evaluation completion rules	Team is Complete when counted evaluations ≥ the round's required evaluations. Otherwise Incomplete. A team with 0 counted evaluations is Unranked. Approval requires every non-disqualified, submitting team to be Complete, or to have its missing assignments excluded with a reason (FR-031) leaving ≥ 1 counted evaluation, or to be explicitly marked "Not ranked" with a reason	Approve blocked with list	Team with 1 of 2 required evaluations blocks approval until resolved	P0

FR-150	Result ordering	Ranked set = teams with status Submitted/Late (if accepted), not disqualified, ≥ 1 counted evaluation. Sort by T descending (4 dp). Ranking computed separately per track when ranking scope is Per track. Rank uses standard competition ranking (1, 2, 2, 4)	n/a	Teams with T = 80.1234, 80.1234, 79 get ranks 1, 1, 3	P0
FR-151	Tie detection	Teams with equal T (4 dp) in the same ranking scope form a tie group, flagged in review. Ties inside award positions must be resolved via FR-033; others display as shared rank	n/a	Tie at position 3 with 3 award positions blocks approval	P0
FR-152	Configured tie-break criterion	Optional per round, set before judging opens: break ties by higher mean weighted contribution on a chosen criterion; remaining ties still go to FR-033	n/a	Setting cannot change after rubric lock	P1
FR-153	Rubric lock and change Ⓐ	Rubric locks when the first evaluation in the round is submitted. While Locked: description text may be edited (logged, no effect on scores). Structural edits (criteria, weights, ranges) require "Unlock rubric" with confirmation: "{n} submitted evaluations will be reopened and judges must re-score." Unlock sets all submitted	n/a	After unlock and weight change, no evaluation computed under old weights is counted	P0

		evaluations to Reopened and pre-fills unchanged criteria			
FR-154	Recalculation	T and rankings are derived from current counted evaluations every time they are viewed or approved; never edited by hand. Any change to counted evaluations, disqualification, or assignment status invalidates an existing approval (FR-034)	n/a	Reopening and resubmitting one evaluation changes T accordingly and returns results to Review	P0
FR-155	Assignment change after scoring Ⓐ	Unassigning a judge with a submitted evaluation marks it "Excluded (unassigned)"; it is kept and shown to admins but not counted. Re-assigning the same judge restores it. Reassignment to a new judge creates a new empty assignment	Confirmation required	Excluded evaluation never appears in T	P0
FR-156	Different rubrics across rounds	Each round's rubric and results are independent. Final standings equal the final round's results; cumulative scoring across rounds is P2 (open question OQ-10)	n/a	Changing Round 2 rubric does not affect Round 1 results	P0
FR-157	Judge spread visibility	Review screen shows each judge's mean J across their teams, to help organisers notice harsh or lenient	n/a	Values match evaluation data	P1

	judges. No automatic adjustment

E.3 Required behaviour in specific situations

Situation	Behaviour
Weights do not total 100%	Rubric cannot be activated; judging cannot open
A criterion has no score	Evaluation stays Draft; Submit blocked
Judge submits only some criteria	Rejected; same as above, including via API
Judge submits an invalid score	Rejected whole; field error; nothing stored
Two teams have identical T	Tie group flagged; resolved per FR-033 if inside award positions
Different judges score the same team	Mean of counted J; no normalisation; judge spread visible (P1)
Judge's assignment changes after scoring	FR-155; approval invalidated if one existed
Submitted evaluation needs correction	FR-100 reopen flow; prior version retained; recalculation and approval invalidation
Teams have different numbers of counted evaluations	Allowed only after explicit admin exclusion with reason; reason shown in approved snapshot

6.F Module F: Google Sheets integration

Recommendation: one-way, snapshot push from the platform to a platform-created spreadsheet, automatic with a short delay, plus "Sync now". CSV export (FR-039) is the P0 fallback; the Sheets push is P1 and must not delay P0.

Option	Pros	Cons	Verdict
CSV export only	Trivial; no OAuth	Organisers re-export manually; stale copies circulate	P0 baseline
One-way snapshot push (rewrite whole managed tab)	No duplicate rows by design; immune to organisers re-sorting rows; simple retry	Organiser edits in managed tabs are overwritten	**Recommended for V1**
One-way incremental row upsert	Fewer API writes	Needs row matching; breaks when rows are sorted, deleted or inserted by humans	Rejected
Bidirectional sync	Organisers edit anywhere	Conflict resolution, validation bypass, audit gaps	Not in V1 (non-goal)

Source-of-truth model. The platform database is canonical. The sheet is a read copy, labelled with "Last synced at {time}". Edits made in the sheet are never imported. This limitation appears in three places: the connection screen, a frozen banner row in each managed tab ("Managed by {platform}. Changes here are overwritten on the next sync. Add your own tabs for notes."), and the organiser help page.

F.1 Sheet structure

Tab	Columns (in order)

Participants	Participant ID, Full name, Email, Phone, Institution, Course, Year, Name on certificate, Team ID, Team name, Team role, Registration status, Registered at, Check-in status, custom fields (by label), Last updated
Teams	Team ID, Team name, Leader name, Leader email, Member count, Members (names, semicolon-separated), Complete (Yes/No), Track, Problem statement, per round: submission status and submitted at, Disqualified, Check-in status, Last updated
Submissions (P1)	Team ID, Team name, Round, Version, Status, Submitted at, Late by (min), Project title, link columns

Scores, judge comments and feedback are never pushed to Sheets in V1, so sheet sharing cannot leak evaluation data.

F.2 Requirements

ID	Feature	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-170	Google account connection Ⓐ	Core Admin clicks Connect → Google OAuth consent. Requested scope: access only to files the app creates (Drive per-file scope), not the whole Drive. Admin must tick: "The sheet will contain participants' personal data. I am responsible for who it is shared with." Tokens stored encrypted; never shown in UI or logs	Consent denied: "Connection cancelled – nothing was changed"	Connection shows the Google account email and connected-by admin	P1

FR-171	Spreadsheet creation	On first connect the platform creates a spreadsheet named "{Event name} – Participants & Teams" in the connected account and stores its ID. Selecting an existing spreadsheet is P2	Creation fails: error with retry; connection stays "Connected, no spreadsheet"	Spreadsheet link opens the created file	P1
FR-172	Initial export	Immediately after creation, a full push of all managed tabs	Same as FR-175	Row counts equal platform counts	P1
FR-173	Sync triggers	Registration, profile, team, membership, track/statement, check-in and submission status changes enqueue a sync for the event. Jobs are coalesced: at most one queued and one running job per event; a job runs no sooner than 60 s after the latest change and no later than 5 minutes after the first queued change (targets). "Sync now" enqueues immediately	n/a	50 registrations in one minute produce at most 2 sync jobs	P1
FR-174	Duplicate prevention and matching	Each job reads current platform data and rewrites each managed tab completely (clear, then write). Managed tabs are identified by stored sheet tab ID, not name, so renaming a tab does not create a new one. Each row carries	Managed tab deleted by organiser: recreated on next sync with a notice	Running Sync now 3 times produces identical tab contents with no duplicate rows	P1

		Participant ID or Team ID for human matching			
FR-175	Retry behaviour	Failed jobs retry after 1, 2, 4, 8 and 16 minutes, then become Failed (final). Rate-limit responses back off the same way. A newer queued job supersedes a failed older one	Final failure: dashboard and integration page show red status with message	No retry loop continues beyond 5 attempts	P1
FR-176	Sync status	Shows: connection status, spreadsheet link, last successful sync time, last attempt time and result, queued/running job, last 20 job results	n/a	"Last successful sync" never updates on a failed job	P1
FR-177	Error handling	Plain-language mapping: token revoked → "Google access was removed. Reconnect to resume syncing."; file deleted or access lost → "The spreadsheet can't be found. Create a new one or reconnect."; quota → "Google is limiting requests; retrying automatically." Technical detail available to admins under "Details"	n/a	Each mapped error is reproducible in a test environment	P1

FR-178	Disconnect and reconnect Ⓐ	Disconnect stops all jobs and deletes stored tokens; the spreadsheet stays in Google untouched. If the connecting admin leaves the event, the connection pauses with "Needs reauthorisation" until another admin reconnects. Reconnect offers: resume with the same spreadsheet (if still accessible) or create new	n/a	After disconnect, a registration change creates no job	P1
FR-179	Data consistency	Each job writes "Last synced at {time} (data as of {time})" in the banner row. Partial write failure marks the job Failed and the next job rewrites the full tab	n/a	No state exists where status says Succeeded but a tab was not written	P1
FR-180	Truthful status	A sync is marked Succeeded only after Google confirms all writes for that job	n/a	Simulated failure on the second tab leaves status Failed	P1
FR-181	Export fallback	When Sheets is disconnected or failing, integration page and dashboard card link to CSV export (FR-039)	n/a	CSV available regardless of integration state	P0

6.G Module G: Google Drive integration and file storage

Recommendation: hybrid. Native platform storage is canonical for all uploaded files; judges and participants access files only through the platform. Google Drive is an optional one-way archive mirror of submitted versions into the organiser's Drive (P1).

Option	Operational complexity	Verdict
Drive as primary store	High. Judges and participants would need Google accounts or public links; per-file sharing must track assignments and removals; Drive quotas and outages would block submissions near deadlines	Rejected for V1
Native storage only	Low. Platform controls access with its own permissions	P0
Native + Drive archive mirror	Medium. Mirror runs after submission, so Drive problems never block a team or a judge	**Recommended (mirror P1)**

The access rule is simple: **nobody gets a file through Drive sharing.** The platform never shares Drive files with judges or participants; mirrored files are private to the connected organiser account. This removes the need to grant or revoke Drive permissions when assignments change.

ID	Feature	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-190	Native file storage	Uploaded files stored in private platform storage under random keys, linked to event, round, team, submission version and field. Metadata: original filename, size, content type, checksum, uploaded by, uploaded at	Storage write failure: upload fails visibly (FR-066)	Files are never reachable by a guessable URL	P0

FR-191	Drive connection Ⓐ	Same OAuth pattern and per-file scope as FR-170 (one Google connection may serve both Sheets and Drive). Admin chooses whether to enable mirroring	Consent denied	Connection status visible on integration page	P1
FR-192	Folder structure	Platform creates: "{Event name} ({slug})" › "Round {n} – {round name}" › "{Team ID} – {Team name}". Folder IDs are stored; later team renames do not rename folders in V1	Folder creation fails: job retries per FR-175	Two teams with the same display name never share a folder (Team ID prefix)	P1
FR-193	Mirror on submission	When a submission version is submitted (never drafts), each file is copied to the team folder as "v{version}*{field label}*{original name}". Links (repository, video, live) are not copied; a "links.txt" file lists them	Failure marks the file Not mirrored; submission success is unaffected	A resubmission creates new v{n+1} files and keeps older ones	P1
FR-194	Mirror metadata and duplicate handling	Each mirror job is keyed by (submission version, field). If a Drive file ID is already stored for that key, the job is skipped. Per-file mirror status: Pending / Mirrored / Failed, with Drive link (admins only)	n/a	Retrying a completed job creates no duplicate file	P1

FR-195	Judge and participant file access	Files are served only after a server permission check (Section 8), via signed links that expire after 10 minutes (target). Judges see files for assigned teams in rounds where judging is Open or Closed; participants see their own team's files; admins see all	Expired link: viewer refreshes it automatically if still authorised	Removing a judge's assignment stops new links being issued immediately	P0
FR-196	Deletion and replacement rules	Platform never deletes files from Drive. Replacing a file in a draft discards only the unsubmitted draft file; submitted versions are immutable. Native files are retained until the event's data is deleted (Section 11)	n/a	No file belonging to a submitted version can be deleted by a participant	P0
FR-197	Failed transfer recovery	Retry schedule as FR-175; "Retry failed" button; "Backfill" mirrors every submitted version not yet mirrored	n/a	After a simulated outage, backfill mirrors all missing files once	P1
FR-198	Disconnect behaviour Ⓐ	Disconnect cancels pending mirror jobs, keeps already mirrored files in Drive, and keeps all native files. Reconnect offers backfill	n/a	Judges' access is unaffected by Drive disconnection	P1

FR-199	Inaccessible external links	The platform does not verify access to links teams paste (e.g. a Drive or YouTube link set to private). Judges can "Report link not accessible" (FR-094); admins see the report and can contact the team	n/a	Report appears in admin submission view for that team	P0

6.H Module H: Feedback

V1 has exactly two event feedback forms (participant and judge) plus controlled release of judges' team-facing comments. **Minimum workflow:** admin opens the default form (optionally edited) → users submit once → admin views aggregates and exports → after results are published, admin releases judges' team comments.

Default participant form: Overall experience (rating 1–5, required); Organisation and communication (1–5); Ease of using the platform (1–5, required); Clarity of judging criteria (1–5); What worked well (long text); What should improve (long text); Would you participate again (Yes / No / Maybe).

Default judge form: Overall experience (1–5); Clarity of the rubric (1–5, required); Ease of the evaluation workspace (1–5, required); Was the time sufficient (Yes / No); Suggestions (long text).

ID	Feature	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-210	Feedback forms Ⓐ	One participant form and one judge form per event, created from the defaults. Question types: rating 1–5, single choice (≤ 10 options), short text, long text; required flag; ≤ 15 questions. States: Draft / Open / Closed. Once a form has responses, questions	Edit blocked with explanation	Editing wording of an existing question is logged	P0

		cannot be deleted or change type; new questions must be optional			
FR-211	Availability	Admin opens a form manually and may set an automatic close time. The dashboard suggests opening forms once final results are published. Eligible respondents: active (not removed) participants for the participant form; active judges for the judge form	Not open: "Feedback isn't open yet"; closed: "Feedback has closed"	Ineligible user receives not found	P0
FR-212	Response submission and duplicate prevention	Required questions validated. One response per user per form (database constraint); responses cannot be edited after submission. Confirmation shown	Second attempt: "You've already submitted feedback. Thank you!"	Double submit stores one response	P0
FR-213	Confidential vs identified	Per form setting, chosen before opening and locked once responses exist. Confidential (default for participant form): organiser views and exports never show identity; identity is stored only to prevent duplicates. Form text states: "Organisers won't see your name with your answers. Free-text answers may still identify you." The word "anonymous" is not used, because identity is stored. Identified (default for judge form): name shown	n/a	No admin screen or export reveals identity for a Confidential form	P0

FR-214	Organiser review	Response rate (responses / eligible), per-question averages and distributions for ratings and choices, list of text answers, CSV export	No responses: empty state with response rate 0	Averages match exported raw data	P0
FR-215	Access restrictions	Only Core Admins can view responses. Judges, staff and participants cannot view any responses, including their own after submission (they see a receipt only)	n/a	Judge request for responses returns not found	P0
FR-216	Release of judge comments to teams Ⓐ	Source: "Feedback for the team" field of counted evaluations. Admin reviews comments for a round and may hide an individual comment with a reason (cannot edit text). "Release team feedback" is possible only after that round's results are published. Teams see comments labelled "Judge 1, Judge 2…" without names. Private notes are never shared. Scores are shown only if the result setting "show scores" is on	Release before results published: blocked	Team A never sees Team B's comments; hidden comment never appears	P1
FR-217	Feedback reminders	Email reminder to non-respondents, once, triggered by admin	n/a	Only non-respondents receive it	P1

6.I Module I: Certificate management

Minimum viable workflow: upload a background image per certificate type → place fields → preview → generate for eligible people → review → release → participants download PDFs. Corrections regenerate the same certificate ID.

ID	Feature	Behaviour and validation	Failure states	Acceptance criteria	Pri
FR-230	Certificate templates Ⓐ	Types: Participation and Winner (Judge and Staff appreciation: P1). Template: background image (PNG/JPG, ≤ 10 MB, A4 landscape; ≥ 2480×1754 px recommended) with signatures and logos baked in. Placeable fields: recipient name, event name, team name, award text, date, certificate ID. Each field: position, font (choice of 3 bundled fonts), size, colour, alignment, max width (text shrinks to fit, minimum 60% of set size, then wraps to 2 lines)	Image too small: warning; wrong type: error	Preview with the longest real participant name stays inside its box	P0

FR-231	Eligibility rules	Participation default: active member of a non-disqualified team that has a submitted version in Round 1. Alternative rule (per event): checked in (FR-126). Winner: members of teams at award positions in the published final results; award text configured per position and scope (e.g. "First Place – FinTech Track"). Winners also receive Participation (configurable). Admin can include or exclude individuals with a reason. Membership is evaluated at generation time	Winner generation before final results published: blocked	Removed or withdrawn participants are never eligible	P0
FR-232	Generation and duplicate prevention Ⓐ	Background job creates one certificate per (event, user, type, award) — database-unique — with a random certificate ID (e.g. CERT-7F3K-9Q2M) and a PDF. Status: Generated (not visible to participants). Re-running generation only adds newly eligible people and flags holders who are no longer eligible	Per-certificate failure listed with retry; job continues for others	Running generation twice creates no duplicates; 500 certificates complete within the target in Section 11	P0
FR-233	Preview and review	List with name, type, award, status (Generated / Needs regeneration / Released / Revoked / Flagged ineligible). Open any PDF. Admin confirms review	n/a	Flagged and Needs-regeneration items block release of that type until	P0

		before release (checkbox "I have reviewed a sample")		resolved or acknowledged	
FR-234	Release Ⓐ	Release per type (or per selected certificates). Released certificates become downloadable. Email notice to recipients: P1	n/a	Unreleased certificates never appear for participants	P0
FR-235	Download	Participant sees the event's certificates and a global "My certificates" page across events. PDF filename: "{Event slug}*{Type}*{Name}.pdf"	Not released: "Certificates will be available after the organisers release them"	Participant can download only their own certificates	P0
FR-236	Name correction Ⓐ	Before generation: participant edits "name on certificate" (FR-056). After generation: participant uses "Request name correction" with the corrected name; admin approves; certificate regenerates with the same ID and an incremented version; old PDF stops being served. If already released, the corrected version is released automatically	Declined: participant sees reason	Corrected PDF shows new name and same certificate ID	P0

FR-237	Revocation and ineligibility Ⓐ	Admin revokes with a reason (e.g. disqualification after release). Status Revoked; download removed; participant sees "This certificate was withdrawn by the organisers." Reversible by admin	n/a	Revoked certificate cannot be downloaded via an old link	P0
FR-238	Public verification page	Anyone enters a certificate ID to confirm name, event and type	n/a	n/a	P2
FR-239	Judge and staff appreciation certificates	Same mechanism with eligibility = active judges or staff	n/a	n/a	P1

7. End-to-end user flows

Each flow lists the user action, system response with validation, and the failure or alternative path; success is the end state of each step unless stated. Requirement IDs point to the detailed rules.

7.A Organiser flow

Entry point: sign-in page or My Events.

#	User action	System response and validation	Failure or alternative path

A1	Signs in	Verifies credentials; shows My Events (FR-001, FR-002)	Wrong password, lockout, unverified email
A2	Creates event	Validates required fields; creates Draft event with Round 1 and default guidebook sections (FR-003)	Validation errors; slug conflict suggestion
A3	Configures eligibility, team size, registration fields	Validates min ≤ max, field limits (FR-013–FR-015)	Invalid range; too many custom fields
A4	Creates tracks and problem statements (optional)	Validates unique names, attachments (FR-010, FR-011)	Skipped entirely for a single-track event
A5	Sets round deadlines and submission form	Validates date ordering; applies default form for event type (FR-012)	Ordering error
A6	Configures rubric (moved earlier than the brief's order; Recommendation)	Copies generic or builds custom rubric; activation requires weights = 100 (FR-023–FR-025)	Activation blocked; can stay Draft until judging opens
A7	Edits and publishes guidebook	Draft autosave; publish creates version and PDF (FR-021, FR-022)	PDF failure: web guidebook still published
A8	Publishes event	Checks publish prerequisites (FR-005)	Blocked with checklist

A9	Opens registration	Registration status Open; landing page shows Register	Scheduled open time instead
A10	Reviews participants and teams	Lists, filters, corrections (FR-016–FR-020)	Conflicts resolved via FR-018
A11	Closes registration	Self-service team changes stop; incomplete teams flagged	Admin edits remain possible
A12	Invites and assigns judges	Invites; assignment per round (FR-026, FR-027)	Role conflict; conflict-of-interest reassignment
A13	Monitors submissions	Status counts; extensions (FR-028)	Missing submissions; late flags
A14	Opens judging	Allowed after deadline + grace and Active rubric (FR-029)	Blocked with reason
A15	Monitors evaluations	Progress; missing evaluations (FR-030, FR-031)	Reassign, exclude with reason, correction requests
A16	Closes judging; reviews scores and ties	Computed ranking, flags (FR-032, FR-151)	Ties at award positions must be resolved (FR-033)

A17	Approves results	Preconditions check; immutable snapshot (FR-034)	Blocked checklist; later changes invalidate
A18	Publishes results (and shortlist for multi-round)	Participants see results per visibility settings (FR-035, FR-044)	Unpublish with reason
A19	Opens feedback; releases team comments	FR-210–FR-216	Release blocked until results published
A20	Generates, reviews, releases certificates	FR-230–FR-235	Per-certificate failures; corrections; revocation
A21	Exports records; concludes event	CSV exports; Concluded state (FR-039, FR-008)	Large exports run in background

Dependencies and parallel work. A2 must precede everything. A3–A7 can run in any order and in parallel, but A8 requires A3 (team size) and A5. A6 must be complete before A14, and should be complete before A9 so the guidebook's judging section matches the rubric. Judge invitations (A12) can run in parallel with registration (A9–A11); assignments are finalised after A11. A14 requires the submission deadline to pass. A17 requires A16; A18 requires A17. Winner certificates (A20) require A18 for the final round; participation certificates require only Round 1 submissions to be locked. Feedback (A19) and certificates (A20) can run in parallel. Staff tasks can be created any time after A8.

7.B Participant / team flow

Entry point: shareable event link.

#	User action	System response and validation	Failure or alternative path

B1	Opens event link	Public landing page (FR-050)	Draft/archived: "Event not found"
B2	Reads event information and guidebook	Published guidebook only (FR-060)	Not yet published: placeholder
B3	Clicks Register; signs in or signs up	Account and email verification (FR-001)	Registration not open
B4	Completes registration form	Validates required fields and declarations; one registration per event (FR-051)	Already registered → dashboard
B5a	Creates a team	Unique name; Team ID and join code issued (FR-052)	Name taken
B5b	Or joins with a code	Atomic size check (FR-053)	Full team, already in a team, invalid code, rate limit
B6	Leader selects track / problem statement	Lock at Round 1 deadline (FR-061, FR-062)	Statement full (P1) or not yet visible
B7	Reads submission requirements	FR-064	n/a
B8	Prepares project offline	n/a	n/a

B9	Uploads files and adds links; draft saves	Type/size validation; URL validation (FR-066–FR-069)	Upload failure with retry; other fields kept
B10	Submits	Validates required fields, team completeness, deadline (FR-065, FR-071)	Incomplete; team below minimum; deadline passed
B11	Receives confirmation	On-screen receipt with server timestamp; email to all members (FR-070)	Email failure does not undo submission
B12	Edits before deadline (optional)	New draft; must resubmit (FR-072)	Unsubmitted changes warning
B13	Checks status	Locked after deadline (FR-073)	"Not submitted" if only a draft existed
B14	Views results and shared comments	After publication/release (FR-074, FR-216)	Placeholder until published
B15	Provides feedback	One response (FR-212)	Not open; already submitted
B16	Downloads certificate	After release (FR-235)	Not released; revoked

Team and deadline edge cases. Joining a full team fails cleanly. A participant can never be in two teams. Leaders must transfer leadership before leaving. Membership freezes at registration close; later changes are admin-only. A team below minimum size cannot submit without an admin override. Drafts are not submitted automatically. The official version is the last submitted one at the deadline. A team-specific extension reopens editing only for that team.

7.C Judge flow

Entry point: invite email or sign-in.

#	User action	System response and validation	Failure or alternative path
C1	Signs in	Judge membership verified (FR-090)	Expired invite
C2	Opens assigned event and round	Round list with judging status (FR-091)	Judging not open: read-only list
C3	Views assigned teams	Only assigned teams (FR-092)	No assignments: empty state
C4	Selects a team	Opens workspace; conflict option available (FR-093, FR-097)	Declares conflict → scoring disabled, admin alerted
C5	Reviews team and project info	FR-093	Missing optional fields hidden
C6	Opens submission files	Authorised signed links (FR-094, FR-195)	File fails to load → "Report missing file"; inaccessible external link → report
C7	Scores each criterion	Integer range validation; live weighted score; autosave (FR-095, FR-096)	Invalid value; autosave retry

C8	Adds feedback	Team comment and private note	Required comment missing
C9	Submits evaluation	Confirmation dialog; idempotent submit (FR-098, FR-099)	Judging closed; assignment removed; already submitted
C10	Sees confirmation	Status Submitted; progress updated (FR-101)	n/a
C11	Proceeds to next team	"Next team" goes to next Not started team	All done: "All assigned evaluations submitted"

Missing files: judge may still score; the report is visible to admins, who may extend for that team (which reopens the submission, pauses that team's assignments as "Awaiting resubmission", and reopens any already submitted evaluations once the team resubmits) or accept as is. **Incomplete evaluations** remain Drafts and are never counted. **Accidental submission:** the judge requests a correction (FR-100); nothing changes until an admin approves.

7.D Helping staff flow

Entry point: invite email or sign-in, usually on a phone.

#	User action	System response and validation	Failure or alternative path
D1	Signs in	Staff membership verified (FR-120)	Expired invite
D2	Views assigned tasks	Only own tasks (FR-122)	No tasks: empty state

D3	Opens a task; sets In progress	FR-123	n/a
D4	Looks up the team	Limited card, no emails (FR-124)	No match → report issue
D5	Verifies members against ID	Marks ID verified (FR-125)	Mismatch → report issue (D7)
D6	Checks in members	Idempotent check-in (FR-126)	Network failure: toggle reverts with retry
D7	Reports an issue if needed	Issue in admin queue (FR-128)	Missing description
D8	Escalates unresolved issue	Severity Urgent; pinned on admin dashboard (FR-129)	n/a
D9	Marks task complete	Confirmation; admin sees completion (FR-130)	Blocked status with note instead

7.E Data synchronisation flow

#	Step	Failure handling	User-facing result
E1	Team registers	Validation failure: nothing stored	Field errors
E2	Platform validates and stores the	Transaction fails: nothing stored, no sync queued	"Something went wrong – your registration was not saved. Try again."

	canonical record in one transaction		
E3	Sheets sync job enqueued (if connected)	Queue failure: record remains saved; integration marked "Sync pending – queue error"; next change or Sync now re-enqueues	Participant sees success; admin sees sync warning
E4	Job rewrites managed tabs	Google error: retry schedule (FR-175); last successful time unchanged	Admin sees Failed/retrying with message
E5	Team uploads file and submits	Upload failure: field error, submission not possible until resolved	Participant sees retry
E6	Submission stored natively with metadata and version	Storage failure: submit fails with clear message	"Your submission was not received. Try again."
E7	Drive mirror job enqueued (if enabled)	Mirror failure never affects E6; per-file status Failed; retry/backfill	Admin sees mirror status; participant unaffected
E8	Judge opens submission after judging opens	Permission check; signed link from native storage	Judge unaffected by Sheets/Drive state

Rule: every integration status displayed is derived from confirmed job results; no step reports success before the external service confirms it (FR-180).

8. Role-based access control

Permissions are granted per event through a single membership role; every request is authorised on the server against the event that owns the requested record, never against what the interface currently shows.

8.1 Permission matrix

Legend: **Full** = all records in the event; **Read** = read-only; **Assigned** = only records assigned to the user; **Own** = only the user's own record or own team; **None** = no access (request returns not found).

Permission	Core Admin	Helping Staff	Participant	Judge
View event (published info, guidebook)	Full (incl. drafts)	Read (published)	Read (published)	Read (published)
Create events	Any verified user can create; creator becomes Core Admin	Same (global)	Same (global)	Same (global)
Edit event settings, tracks, rounds, guidebook	Full	None	None	None
Manage participants (view/edit records)	Full	Read, limited fields (Module D)	Own	None
Edit team records and membership	Full	None	Own team (leader; until registration close)	None

Record check-in and ID verification	Full	Full within event	None	None
View submissions	Full	None	Own team	Assigned (judging Open/Closed)
Upload submissions	None (open question OQ-12)	None	Own team (before deadline)	None
Edit submissions	None; can extend/unlock	None	Own team (before deadline)	None
Unlock or extend a submission	Full	None	None	None
View judging rubrics	Full	None	None (guidebook text only)	Assigned rounds, Read
Edit judging rubrics	Full	None	None	None
Assign judges	Full	None	None	None
Enter scores	None	None	None	Assigned

Edit submitted scores	None (reopen only)	None	None	Own, only when reopened
View judge comments	Full (team comments and private notes)	None	Own team, released team comments only, no judge names	Own
View aggregate scores and rankings	Full	None	Own team's published rank/score if enabled; published winners	None
Approve results	Full	None	None	None
Publish results	Full	None	None	None
Manage feedback forms	Full	None	None	None
View feedback responses	Full (identity hidden for Confidential forms)	None	None	None
Submit feedback	None	None	Own	Own
Create and generate certificates	Full	None	None	None

Release and revoke certificates	Full	None	None	None
Download certificates	Full	Own (P1)	Own	Own (P1)
Export participant data (CSV)	Full	None	None	None
Manage Google integrations	Full	None	None	None
View audit history	Full	None	None	None
Manage staff tasks	Full	Assigned (status only)	None	None
Report and view issues	Full	Own reports	None	None
Post announcements	Full	Read (staff/everyone audience)	Read (participant/everyone audience)	Read (judge/everyone audience)

8.2 Multiple roles

1. A user holds **at most one role per event**, enforced by a unique constraint on (event, user) in memberships. Invitations or registrations that would create a second role in the same event are rejected with a message naming the existing role.

2. A user can hold different roles in different events (e.g. Judge in Event A, Participant in Event B). Permissions never carry across events: a Judge in A gets nothing in B from that role.
3. Core Admin includes every staff capability, so organisers never need a second role. A judge who also wants to help at the check-in desk is not supported in V1 (open question OQ-13).
4. Every authorisation check resolves the event from the requested record itself (e.g. submission → team → event), so a crafted request cannot borrow a role from another event.

8.3 Additional rules

- Records outside a user's scope return "not found" rather than "forbidden", to avoid revealing that a record exists. Denied attempts are logged.
- Signed file links are issued only after the permission check and expire within 10 minutes (target).
- Platform support staff are not a product role. Any direct database access by the development team during the pilot must be logged outside the product (operational policy).

8.4 Removal from an event

Role removed	Effect
Participant	Membership status Removed; leaves team (leader must be reassigned first if others remain); all event requests denied from the next request; excluded from certificates; registration data retained for admins and audit
Judge (deactivated)	Access denied from next request; submitted evaluations remain counted unless an admin excludes them; drafts are discarded from consideration; unsubmitted assignments appear in missing evaluations for reassignment
Helping Staff	Access denied; their open tasks become Unassigned and appear on admin dashboard; check-in records they made remain
Core Admin	Access denied; owner cannot be removed; integration connections they created pause until reconnected (FR-178)

In all cases the user's account and other event memberships are unaffected. Previously downloaded files cannot be recalled; this is stated in the admin confirmation dialog.

9. Data model and entity relationships

The model centres on Event: almost every entity carries an event ID, which is what makes per-event permission checks and duplicate prevention enforceable in the database. This is conceptual; it is not a production schema.

9.1 Entities

Entity	Purpose	Important fields	Relationships	Uniqueness / integrity	V1
User	Global account	id, email, name, password hash or sign-in method, email_verified	Has many EventMemberships	email unique (case-insensitive)	Essential
Event	A competition	id, slug, name, type, mode, timezone, dates, status, registration status, team min/max, settings	Has many of most entities below; owner = User	slug unique	Essential
EventMembership	User's role in an event	id, event_id, user_id, role (admin/judge/staff/participant), status (invited/active/removed), invited_by	Belongs to Event and User	(event_id, user_id) unique → one role per event	Essential
Registration	Participant's form	id, membership_id, fixed fields, custom answers (key–value), declarations accepted, status	1:1 with participant	one per participant membership	Essential (could

	answers		EventMembership		merge into membership)
RegistrationField	Custom field definition	id, event_id, label, type, options, required, hidden, visible_to_staff	Belongs to Event	label unique per event	Essential
CompetitionTrack	Category	id, event_id, name, description, order	Belongs to Event; has many ProblemStatements, Teams	(event_id, name) unique	Essential (optional use)
ProblemStatement	Brief or case	id, event_id, track_id (nullable), title, description, attachment, visible_from, cap	Belongs to Event and optional Track	(event_id, title) unique	Essential (cap, visible_from: P1)
CompetitionRound	Stage with deadline	id, event_id, order, name, opens_at, deadline, grace_min, late_policy, judging_deadline, judging_status, ranking_scope, required_evaluations	Belongs to Event; has one Rubric, many SubmissionFormFields, JudgeAssignments	(event_id, order) unique	Essential

SubmissionFormField	Round's submission field definition	id, round_id, type, label, help, required, order, file limits	Belongs to Round	(round_id, label) unique	Essential
Team	Competing unit	id, event_id, team_code (T-0042), name, leader_member_id, track_id, problem_id, join_code, status (active/dissolved/disqualified)	Belongs to Event; has many TeamMembers, Submissions	(event_id, team_code) unique; (event_id, lower(name)) unique among active; join_code unique per event	Essential
TeamMember	Participant in team	id, team_id, membership_id, event_id, role (leader/member), joined_at, left_at	Belongs to Team and EventMembership	(event_id, membership_id) unique among active rows → one team per event	Essential
RoundParticipation	Team eligible for a round	round_id, team_id, source (all/shortlist)	Links Round and Team	(round_id, team_id) unique	Simplify: implicit for Round 1; needed for multi-

					round (P1)
Submission	A team's entry for a round	id, round_id, team_id, status, current_draft_version_id, last_submitted_version_id, extension_deadline	Belongs to Round and Team; has many SubmissionVersions	(round_id, team_id) unique	Essential
SubmissionVersion	Immutable submitted snapshot (drafts mutable)	id, submission_id, version_no, state (draft/submitted), submitted_at, submitted_by, late_minutes, field values	Belongs to Submission; has many SubmissionFiles	(submission_id, version_no) unique	Essential
SubmissionFile	Stored file	id, version_id, field_id, storage_key, original_name, size, content_type, checksum, drive_file_id, mirror_status	Belongs to SubmissionVersion	(version_id, field_id) unique in V1 (one file per field)	Essential (Drive fields: P1)
Guidebook / GuidebookVersion	Rules content	guidebook: event_id, draft sections; version: version_no, sections snapshot, published_at, published_by, pdf_key, change_note	Event has one Guidebook; Guidebook	(event_id) unique; (guidebook_id, version_no) unique	Essential

			has many versions		
Announcement	Event update	id, event_id, audience, title, body, author, created_at, edited_at	Belongs to Event	n/a	Essential
Rubric	Scoring definition	id, round_id, name, state, team_comment_mode, source_template	One per Round	(round_id) unique	Essential
RubricCriterion	Criterion	id, rubric_id, name, description, weight, min, max, order	Belongs to Rubric	(rubric_id, name) unique; weights sum to 100 on activation	Essential
JudgeAssignment	Judge × team × round	id, round_id, team_id, judge_membership_id, status (active/conflict/excluded/unassigned/not_required/awaiting_resubmission), exclusion_reason	Links Round, Team, judge membership; has 0..1 Evaluation	(round_id, team_id, judge_membership_id) unique	Essential
Evaluation	A judge's scoring of a team	id, assignment_id, status (draft/submitted/reopened), weighted_score (4 dp), team_comment, private_note, submitted_at, version_no, rubric_version	Belongs to JudgeAssignment; has many CriterionScores; history in	(assignment_id) unique	Essential

			Evaluation Version		
CriterionScore	One criterion's score	evaluation_id, criterion_id, score (integer)	Belongs to Evaluation and RubricCriterion	(evaluation_id, criterion_id) unique	Essential
EvaluationVersion	Prior submitted states after reopen	evaluation_id, version_no, scores, comments, submitted_at	Belongs to Evaluation	(evaluation_id, version_no) unique	Essential (audit)
ResultSnapshot	Approved, immutable ranking	id, round_id, version_no, entries (team, score, rank, scope), tie resolutions, exclusions, approved_by/at, published_at, status (approved/invalidated/published)	Belongs to Round	(round_id, version_no) unique	Essential
CheckIn	Attendance mark	event_id, member_id, checked_in_at, by_membership_id, id_verified, note	Belongs to TeamMember	(event_id, member_id) unique	Essential
StaffTask	Assignment	id, event_id, title, description, type, due_at, location, status, assignees, linked teams	Belongs to Event	n/a	Essential

	for staff				
IssueReport	Staff-raised issue	id, event_id, reporter, category, severity, team_id, description, status, resolution_note	Belongs to Event	n/a	Essential
Feedback Form / Feedback Question	Form definition	form: event_id, audience (participant/judge), state, confidential, close_at; question: type, text, options, required, order	Event has ≤ 2 forms	(event_id, audience) unique	Essential
Feedback Response	One submission	form_id, respondent_membership_id, answers, submitted_at	Belongs to Form	(form_id, respondent_membership_id) unique	Essential
Certificate Template	Layout per type	id, event_id, type, background_key, field placements	Belongs to Event	(event_id, type) unique	Essential
Certificate	Issued certificate	id, certificate_code, event_id, membership_id, type, award_key, name_snapshot, version, status, pdf_key, released_at, revoked_reason	Belongs to Event, membership, template	certificate_code unique; (event_id, membership_id, type, award_key) unique	Essential

IntegrationConnection	Google link	id, event_id, provider, google_account_email, encrypted tokens, spreadsheet_id, root_folder_id, status, connected_by	Belongs to Event	(event_id, provider) unique	P1
SyncJob	Integration work item	id, connection_id, kind (sheets_push/drive_mirror), key, status, attempts, next_attempt_at, error, completed_at	Belongs to Connection	(connection_id, key) unique among queued/running → coalescing and idempotency	P1
AuditLog	Immutable change record	id, event_id, actor, action, entity_type, entity_id, before, after, reason, at	Belongs to Event	append-only	Essential

9.2 Key relationships explained

- **Event → Teams (1:n) and Team → Participants (1:n via TeamMember).** A participant reaches a team only through their event membership, so one-team-per-event is a single unique constraint on (event_id, membership_id) for active TeamMember rows.
- **Event → Rounds (1:n, at least one).** A single-round event is simply an event with one round, so no separate code path exists.
- **Team submits for a round:** exactly one Submission per (round, team), holding many versions. Judges and results always use last_submitted_version_id at lock time.
- **Judge assigned to teams:** JudgeAssignment is the join of judge membership, team and round; Evaluation hangs off the assignment, so "one evaluation per judge per team per round" is one unique constraint.
- **Evaluation → CriterionScores (1:n),** one per rubric criterion; a submitted evaluation must have exactly as many scores as the rubric has criteria.

- **Participant → certificates across events:** a User has memberships in many events; each Certificate belongs to one event and one membership, and "My certificates" queries across the user's memberships.

9.3 Integrity and duplicate prevention

1. Uniqueness rules in 9.1 are database constraints, not only interface checks.
2. Team join, team creation, submission, evaluation submission, result approval and certificate generation run in transactions.
3. Write requests that create records (register, submit, evaluate, generate) accept an idempotency key so retries after a network drop do not duplicate.
4. Submitted versions, evaluation versions, result snapshots, guidebook versions and audit entries are never updated in place.
5. Deletion is soft for teams, participants and memberships so that history, audit and results remain explainable.
6. Team codes and certificate codes are never reused, even after dissolution or revocation.

10. Edge cases and failure scenarios

Blocking failures stop the user's action until resolved; recoverable failures let work continue and are fixed later. Integration failures are always recoverable and never block registration, submission or judging.

#	Scenario	Trigger	Expected behaviour	User-facing message	Data consistency requirement	Recovery	Admin needed?	Type
1	Duplicate participant registration	Same user registers twice (double click, two tabs)	Second request returns the existing registration	Redirect to team dashboard	One registration per (event, user)	Automatic	No	Recoverable

2	Same person with two accounts	Different emails used	Both registrations allowed (system cannot know); admin sees possible duplicates by matching name + phone	Admin list badge "Possible duplicate" (P1)	No merging in V1	Admin removes one registration	Yes	Recoverable
3	Duplicate team creation	Two teams with same name	Second blocked by unique name rule	"A team with this name already exists"	(event, name) unique	User picks another name	No	Blocking
4	Team membership conflict	Participant tries to join while in a team	Blocked	Names current team	One team per event	Leave current team first	No	Blocking
5	Team size exceeding limit	Last-slot race; admin reduces max	Atomic join; max reduction blocked	"Team is full" / list of oversize teams	Count ≤ max always	Leader removes member or admin keeps max	Sometimes	Blocking

6	Registration closes before submission deadline	Normal configuration	Team changes locked; submissions continue	"Contact organisers to change your team"	Membership frozen except admin edits	Admin edits via FR-018	If change needed	Recoverable
7	Participant changes after registration	Member drops out after close	Admin removes member; completeness recalculated; certificate eligibility updated	Team sees updated roster	Audit entry	Admin action	Yes	Recoverable
8	Incorrect team details	Wrong track or name	Leader fixes before lock; admin after	n/a	Track change after assignment warns admin	Admin reviews assignments	Sometimes	Recoverable
9	Submission replacement	Team resubmits	New version; previous version retained	"Version {n} submitted"	Old version immutable	n/a	No	Recoverable

10	Late submission	Submit after deadline + grace	Reject or accept-and-flag per policy	Deadline message or "Late by {n} min"	Server time only	Admin extension	Optional	Blocking (Reject)
11	Failed file upload	Network or size error	Field error; other fields saved; submit disabled for missing required file	"Upload failed – retry"	No partial file attached to a submitted version	User retries	No	Recoverable
12	Inaccessible Google Drive link pasted by team	Private link	Judge reports; admin contacts team	"Report link not accessible"	Report recorded	Admin may extend for that team	Yes	Recoverable
13	Sheets sync failure	Token revoked, quota, file deleted	Retry with backoff, then Failed	Plain-language error on integration page	Platform data unaffected; last success time unchanged	Reconnect / Sync now; CSV fallback	Yes	Recoverable

14	Duplicate sync jobs	Burst of changes; double Sync now	Coalesced; full-tab rewrite is idempotent	n/a	At most one queued and one running job per event	Automatic	No	Recoverable
15	Judge assigned to wrong team	Admin error	Admin unassigns; evaluation excluded if already submitted	Judge sees team disappear	Excluded evaluation not counted	Reassign correct judge	Yes	Recoverable
16	Judge submits incomplete scores	Missing criteria	Submit blocked	Lists unscored criteria	No partial evaluation counted	Judge completes	No	Blocking
17	Judge accidentally submits	Wrong score	Correction request → reopen	"Request correction"	Previous version retained; approval invalidated if any	Admin approves reopen	Yes	Recoverable

18	Rubric changes after judging started	Admin edits criteria/weights	Blocked unless Unlock; unlock reopens all submitted evaluations	Confirmation with count	No score computed with mixed rubric versions	Judges re-score	Yes	Blocking until resolved
19	Missing evaluations at deadline	Judge absent	Listed in missing evaluations; approval blocked	Checklist on results screen	No implicit zero scores	Reassign or exclude with reason	Yes	Blocking (approval)
20	Identical team scores	Equal T at 4 dp	Tie flagged; must resolve at award positions	"Tie at position {n}"	Resolution and reason in snapshot	Share or order manually	Yes	Blocking (approval)
21	Publication before evaluations complete	Admin tries to publish early	Impossible: publish requires approval; approval requires completeness	Unmet checklist	Snapshot only from complete data	Complete evaluations	Yes	Blocking

22	Certificate name error	Misspelt name	Correction request → regenerate same ID	"Request name correction"	Old PDF no longer served	Admin approves	Yes	Recoverable
23	User loses access to event	Removed or deactivated	Next request denied; open signed links expire ≤ 10 min	"You no longer have access to this event"	Records retained	Re-invite if mistaken	Yes	Blocking for user
24	Two organisers edit same record	Concurrent saves	Second save rejected (version check)	"Changed by {name} at {time}. Reload."	No silent overwrite	Reload and redo	No	Recoverable
25	Connectivity problems during live event	Venue Wi-Fi drops	Autosave retries; unsaved input kept in tab; non-idempotent actions use idempotency keys; check-in toggles revert on failure	"Offline – changes not saved yet" banner	No duplicate records on retry	Reconnect; organisers keep a CSV export of teams and check-in as paper fallback	Sometimes	Recoverable

26	Google integration unavailable	Google outage	Jobs queue and retry; core features unaffected	Integration warning only	Platform canonical	Automatic resume; backfill	No	Recoverable
27	Team changes track after judges assigned	Before Round 1 deadline	Assignment kept; admin warned	Admin banner	n/a	Admin reassigns if per-track judging	Yes	Recoverable
28	Deadline edited while teams are mid-submission	Admin shortens deadline	Requires reason; announcement prompted; new deadline enforced	Countdown updates on refresh	Audit entry	Admin communicates	Yes	Recoverable
29	Platform outage near deadline	Hosting failure	Admin can extend deadline for all teams with reason	Announcement	Audit entry	Extension	Yes	Recoverable

11. Non-functional requirements

Targets below are proposals to be validated by load testing before the pilot; none are measured benchmarks. Load profile for testing: AS-01 event size, 300 concurrent users, 150 submissions in the final 10 minutes before a deadline.

11.1 Security

ID	Requirement	Test
NFR-S 1	All traffic over HTTPS; HTTP redirects to HTTPS	Scan shows no plain-HTTP endpoints
NFR-S 2	Passwords stored with a slow, salted hash algorithm designed for passwords; never logged	Code review; log inspection
NFR-S 3	Sessions: secure, HTTP-only cookies; idle timeout 7 days for participants, 12 hours for admins (targets); sign-out invalidates server-side	Session reuse after sign-out fails
NFR-S 4	Every API endpoint enforces Section 8 on the server; automated tests cover each matrix cell with one allowed and one denied case	Permission test suite passes 100%
NFR-S 5	Record IDs in URLs are non-sequential; access by guessing returns not found	Enumeration test
NFR-S 6	Google tokens encrypted at rest with keys outside the database; least-privilege per-file scope	Config review
NFR-S 7	Uploaded files: type validated by signature; stored privately; served with download headers except PDFs in the viewer; never executed; malware scanning P1	Upload an HTML file renamed .pdf → rejected

NFR-S 8	CSV exports neutralise formula injection (FR-039)	Cell "=HYPERLINK(...)" exported with leading apostrophe
NFR-S 9	Rate limits: sign-in (FR-001), join code (FR-053), invite and email sending	Exceeding limits returns a clear message
NFR-S 10	Audit log is append-only; no API to edit or delete entries	Attempted update fails

11.2 Performance (targets, p95 at the load profile)

Operation	Target
Admin dashboard load	≤ 2 s
Participant/team list filter or search (1,000 participants)	≤ 1 s
Team dashboard load	≤ 2 s
Judge workspace load (excluding external links)	≤ 2 s; PDF first page visible ≤ 4 s for a 10 MB file
Submission of a form with a 25 MB file on 10 Mbps upload	Upload time dominated by network; server acknowledges within 2 s after upload completes
Result calculation for 250 teams × 3 evaluations × 12 criteria	≤ 1 s

Certificate generation, 1,000 certificates	≤ 10 min in background; progress visible
Sheets sync completion after last change	≤ 5 min (FR-173)

11.3 Reliability

- Committed writes are durable; automated daily database backups retained 30 days, with a restore tested at least once before the pilot. Recovery point objective ≤ 24 h; during the live event window, take an additional manual backup immediately after the submission deadline and after result approval.
- Uploaded files are stored on storage with provider-level redundancy and included in the backup policy.
- Integration jobs survive application restarts (persistent queue) and are idempotent (FR-174, FR-194).
- Score calculation is deterministic: the same inputs always produce identical T and ranks; a test suite covers the worked example and every rule in E.3.
- Uptime target during the pilot event window (registration close to result publication): no unplanned outage longer than 15 minutes; a named on-call developer is available during that window.

11.4 Privacy

- Participant email and phone are visible only to Core Admins (and phone to staff only when enabled). Teammates see each other's email.
- Judges see only name and institution of members of assigned teams.
- Private judge notes are visible only to Core Admins; team-facing comments only after release, without judge identity.
- Evaluation records are visible only to Core Admins and the authoring judge.
- CSV export and Sheets connection are Core Admin only, logged in audit, and the Sheets connection requires a data-responsibility acknowledgement.
- Privacy notice shown at registration stating what is collected, who sees it, and retention.
- Retention (Recommendation, confirm with institution): event data retained 12 months after the event concludes, then deleted or anonymised except certificates (retained so participants can download them). Participants can request deletion of their account; certificates and audit entries are retained in minimised form. Compliance with applicable Indian data protection law should be reviewed by the faculty coordinator or institution (open question OQ-15).

11.5 Accessibility

- Target WCAG 2.1 AA for participant, judge and staff screens.
- Every action reachable by keyboard; visible focus indicator; logical tab order in the scoring form.
- Body text ≥ 16 px equivalent; contrast ≥ 4.5:1.
- Form errors are announced to screen readers, placed next to the field, and summarised at the top.
- Status is never conveyed by colour alone (e.g. "Submitted" text plus icon).
- Staff and participant screens usable at 360 px width.

11.6 Usability

- A first-time organiser can create and publish an event with one round using only in-product help (validated in usability test, target 4 of 5 testers).
- A judge can complete a first evaluation without instructions (target 4 of 5 testers).
- A staff member can check in a team in ≤ 30 seconds from search to confirmation (target, observed).
- Each role's home screen shows its next required action (e.g. "3 teams left to evaluate").
- Destructive or irreversible actions (submit evaluation, approve, publish, release, revoke, unlock rubric) require confirmation that states the consequence.

12. Product metrics and success criteria

Eleven metrics are enough for the pilot; most are collected automatically from event and audit data, and baselines come from the pilot organiser's records of their previous event or are established during the pilot.

Metric	Calculation	Data source	Frequency	Proposed success criterion	Baseline	Auto / manual
Event setup time	Active minutes from create to publish (exclude idle gaps > 15 min)	Event timestamps + observer	Once per event	Completed in one session without developer help	Establish in pilot	Auto + observation

Registration management effort	Admin edits to participant/team records; organiser self-reported hours	Audit log; survey	Weekly during registration	Self-reported hours below previous event	Previous-event estimate	Auto + survey
On-time submission rate	Teams with on-time submitted version / complete teams at registration close	Submissions	At deadline	Establish; no target until baseline exists	Establish in pilot	Auto
Correct submission association	1 − (confirmed mismatches / submissions)	Post-event review	Once	100%	n/a	Manual
Time per evaluation	Median active time per evaluation	Evaluation timestamps	Daily during judging	Establish	Establish in pilot	Auto
Evaluations on time	Submitted evaluations by judging deadline / required evaluations	Evaluations	At judging deadline	≥ 90% (proposal)	Establish	Auto
Manual corrections	Count of reopens, admin record edits after close,	Audit log	Once	Track; investigate every result invalidation	Establish	Auto

	result invalidations					
Integration success rate	Succeeded jobs / (succeeded + final-failed jobs)	SyncJob	Daily	≥ 99% excluding user-caused revocations (proposal)	n/a	Auto
Time to publish results	Result published_at − judging deadline	Rounds, snapshots	Once per round	Below previous-event estimate	Previous-event estimate	Auto + interview
Certificate success rate	Certificates generated without error and not needing correction / certificates generated	Certificates	Once	≥ 98% (proposal)	n/a	Auto
Satisfaction	Mean of "ease of using the platform" (participants) and "ease of evaluation workspace" (judges)	Feedback forms	After event	≥ 4.0 / 5 (proposal)	None	Auto

Observation-based measures (staff check-in time, organiser stress points, workarounds such as a parallel spreadsheet) need a team member observing at the event with a simple log sheet.

13. MVP scope and prioritisation

The brief's 18 required V1 capabilities are all kept, but four are delivered in simplified form: Sheets as one-way push (with CSV as the P0 fallback), Drive as an optional archive mirror, staff as tasks plus check-in, and feedback as two fixed forms. Without these simplifications the scope is too large for a student team's first cycle.

13.1 Feature matrix

Complexity: L = low, M = medium, H = high (relative, for planning only).

#	Feature	Pri	User value	Operational necessity	Dependencies	Complexity	Risk of deferring
1	Role-based interfaces and permissions	P0	Each user sees only their work	Without it, scores and personal data leak	Auth, memberships	M	Cannot run a fair event
2	Event creation and configuration (tracks, rounds, team size, fields)	P0	Replaces setup docs and forms	Everything depends on it	Auth	M	Nothing else works
3	Participant registration	P0	One form, one record	Core data entry	2	M	Back to Google Forms
4	Team and participant records	P0	Clean roster	Submissions and judging	3	M	Mis-attribution returns

				attach to teams			
5	Editable guidebook with PDF export	P0	Rules in one place	Reduces repeated questions	2	M (PDF rendering)	Organisers fall back to Docs; acceptable short-term but brief requires it
6	Structured submissions with deadline enforcement	P0	Proof of submission; fairness	Central to every event	4, native storage	H	Event cannot be judged in-platform
7	Sheets integration (one-way push)	P1; CSV export is P0	Live sheet for faculty	Convenience; CSV covers necessity	3, 4, job queue, OAuth	M–H	Low: CSV export covers the need
8	Drive integration (archive mirror)	P1; native storage is P0	Organisers keep files in their Drive	Not needed to run the event	6, OAuth	M	Low: files remain accessible in platform and via ZIP/CSV
9	Judge assignment	P0	Judges see only their teams	Required for evaluation	4, judge invites	M	Cannot judge

							in-platform
10	Generic and custom rubrics	P0	Consistent criteria	Required for scoring	Rounds	M	Cannot judge in-platform
11	Score calculation	P0	Correct, explainable results	Core trust requirement	9, 10, evaluations	M (logic) / H (correctness risk)	Manual spreadsheets return
12	Results review, approval and publication	P0	Faculty sign-off, clear announcement	Must publish winners	11	M	Results announced off-platform; certificates blocked
13	Basic helping staff workflows (tasks, check-in, issues)	P0 (check-in, tasks); issue alerts P1	Volunteers work from phones without admin access	Needed for in-person events	4	M	Paper lists; acceptable for online events
14	Participant and judge feedback	P0 (forms); comment release P1	Learn what to improve	Brief requires V1	Auth, roles	L–M	Separate Google Form; brief requires it

15	Certificate generation and release	P0	Fast, correct certificates	Brief requires V1	4, 12, templates	H (layout, PDF, corrections)	Mail merge returns
16	Basic operational dashboard	P0	One-glance status	Live-event awareness	Most modules	L–M	Admins hunt across lists
17	CSV export	P0	Data portability and fallback	Fallback for every integration	3, 4, 6, 11	L	No fallback when integrations fail
18	Essential audit history	P0	Defend decisions	Needed for disputes	All write paths	M (must be built in from start)	Retrofitting is expensive

13.2 Explicitly P2 (not in V1)

Public certificate verification page, anonymous judging, optional rubric criteria, half-point scores, judge-bias normalisation, cumulative multi-round scoring, bidirectional Sheets sync, selecting an existing spreadsheet, queue status for staff, duplicate-account merging, payments, QR check-in, public leaderboards, AI features, mentor scheduling, sponsorship CRM.

13.3 Smallest coherent version

The smallest version that runs a real single-round event end to end: authentication and roles; event setup with one round; registration and teams; guidebook with PDF; submissions on native storage; rubric, assignments, judge workspace and scoring; result approval and publication; participant feedback form; participation and winner certificates; dashboard; CSV export; audit log. Staff check-in is added for an in-person pilot. Sheets push, Drive mirror, multi-round shortlisting, judge feedback release and appreciation certificates follow once this works.

14. Implementation phasing

The brief's sequence is revised in one important way: Google integrations move from Phase 2 to the final phase, because they are P1, depend on an external service, and should never block the core path. A short OAuth spike in Phase 2 de-risks them early. Durations are not estimated, because team size, skills, stack and hours are unknown.

Phase 1: Foundations

- **Objective:** a working, permission-safe skeleton holding real event, participant and team data.
- **Features:** FR-001–FR-006, FR-010–FR-020, FR-050–FR-057, FR-039 (participants, teams), audit log infrastructure (FR-046) wired into every write from the start.
- **Dependencies:** technology stack chosen; hosting and email provider available.
- **Deliverables:** event creation and publication, registration, team create/join/manage, admin lists and corrections, CSV export, permission test harness.
- **Acceptance criteria:** AT-01 to AT-04 and AT-19 (Section 16) pass.
- **Testing:** unit tests for constraints (one team per event, team size race), permission tests for every endpoint built so far, usability test of registration and team join with 5 students.
- **Exit criteria:** a mock event with 100 seeded participants and 30 teams can be configured, registered and exported with no manual database edits.

Phase 2: Content and submissions

- **Objective:** teams can read rules and submit reliably.
- **Features:** FR-007, FR-021–FR-022, FR-059–FR-073, FR-028, FR-190, FR-195, FR-196, submission form configuration, submission receipt emails. Spike: Google OAuth connection and one test spreadsheet write (throwaway).
- **Dependencies:** Phase 1 teams and rounds.
- **Deliverables:** guidebook editor and PDF, announcements, submission form with versioning, deadline enforcement, admin submission monitoring and extensions.
- **Acceptance criteria:** AT-05 to AT-07 pass.
- **Testing:** deadline boundary tests (deadline − 1 s, + grace, + grace + 1 s), file type/size tests, a load test with 150 submissions in 10 minutes, PDF output review.
- **Exit criteria:** a rehearsal with real students submits to a mock round with zero mis-attributed or lost submissions.

Phase 3: Judging and results

- **Objective:** produce correct, approved, published results.
- **Features:** FR-023–FR-027, FR-029–FR-035, FR-090–FR-102, FR-140–FR-156.
- **Dependencies:** Phase 2 submissions locked at deadline.
- **Deliverables:** rubric builder, assignment screen, judge workspace, scoring engine, results review, tie resolution, approval and publication.

- **Acceptance criteria:** AT-10 to AT-16 pass; the worked example in E.1 is an automated test.
- **Testing:** scoring engine unit tests for every row of E.3; independent recalculation of a seeded event's results in a spreadsheet by someone who did not write the code; judge usability test with 3–5 people.
- **Exit criteria:** seeded event results match the independent spreadsheet exactly; no permission test failures.

Phase 4: Event-day operations and closing

- **Objective:** support volunteers on the day and close the event properly.
- **Features:** FR-045, FR-120–FR-130, FR-210–FR-215, FR-230–FR-237, FR-043, FR-008.
- **Dependencies:** Phase 3 published results (for winner certificates).
- **Deliverables:** staff tasks, check-in, issues; feedback forms and review; certificate templates, generation, review, release, corrections; operational dashboard.
- **Acceptance criteria:** AT-17, AT-18, AT-20 to AT-23 pass.
- **Testing:** certificate rendering with long names and non-ASCII characters (e.g. Hindi names typed in Devanagari, if fonts support it – open question OQ-16), phone testing of staff screens at 360 px, feedback confidentiality checks on exports.
- **Exit criteria:** full mock event run end to end by people outside the dev team.

Phase 5: Integrations and pilot hardening

- **Objective:** add Google integrations safely and make the pilot dependable.
- **Features:** FR-170–FR-180, FR-191–FR-194, FR-197–FR-198, P1 items chosen by capacity (multi-round shortlist FR-044, comment release FR-216, auto-balance).
- **Dependencies:** Phases 1–4; Google Cloud project configured; Google's app verification requirements for the requested scopes checked early.
- **Deliverables:** Sheets push with status and retries; Drive mirror with backfill; backup/restore drill; load test; runbook for event day.
- **Acceptance criteria:** AT-08, AT-09 pass; integration failure simulations (token revoked, file deleted, quota) behave as specified.
- **Testing:** fault injection for each integration error; full dress rehearsal with the pilot organisers using their real configuration.
- **Exit criteria:** launch-readiness checklist (Section 18.8) complete. If Phase 5 integrations are not ready, the pilot proceeds with CSV export and native storage only.

15. Screen-by-screen requirements

The inventory lists functional behaviour only; layout and visual design are out of scope. Common states apply unless a row says otherwise: **loading** shows skeleton placeholders for the main area with navigation usable; **error** shows a message, a Retry button, and keeps any user input; **no permission** shows "not found" (Section 8.3).

15.1 Shared

Screen	User	Purpose and main components	Actions	Empty / special states
Sign in / sign up / reset	All	Email + password or email link; verification prompt	Sign in, sign up, resend verification, reset	Locked account message; expired link
My Events	All	Event cards with role badge, status, next deadline	Open event, create event, open event link	"You're not part of any event yet" with create and link options
My certificates	Participant (Judge/Staff P1)	Certificates across events	Download	"No certificates yet"

15.2 Core Admin

Screen	Purpose and main components	Information displayed	Actions	Empty / special states	Permission notes
Event overview (dashboard)	Health cards (FR-043), urgent issues, next deadlines	Counts with links	Refresh, open filtered lists	New event: setup checklist with links	Core Admin only

Create / edit event	Event details form; publish and registration controls	Fields of FR-003, status	Save, publish, open/close registration, unpublish	Publish blocked checklist	Owner-only: delete/archive
Setup: eligibility, team size, registration fields	Declarations, min/max, custom field builder	Field list with types	Add, edit, hide, reorder fields	No custom fields: "Only standard fields will be collected"	Locked rules after registrations shown inline
Setup: tracks and problem statements	Lists and forms	Team counts per item	Add, edit, reorder, delete	"No tracks – all teams compete together"	Delete blocked when in use
Setup: rounds and submission forms	Round list, deadlines, policies, field builder	Dates in event time zone	Add round (P1 beyond one), edit, reorder fields	One default round	Deadline change requires reason
Participants and teams	Tabs: Participants, Teams; search, filters, columns	FR-016–FR-020 fields	Open detail, export, bulk select (export only)	"No registrations yet" with event link to share	Contact data visible here only

Team detail	Tabs: Overview, Members, Submissions, Evaluations, Check-in, History	All team data	Edit, add/remove member, transfer leader, disqualify, extend submission	n/a	Evaluations tab admin-only
Guidebook editor	Section list, rich-text editor, version history, preview	Draft vs published state, last published	Edit, add/hide/reorder section, preview PDF, publish	Defaults pre-filled	Autosave indicator always visible
Announcements	List and composer	Audience, date	Post, edit, delete	"No announcements"	n/a
Submission management	Per-round table with status filter	Status, version, time, late minutes, reports	View version, open files, extend, export, bulk ZIP (P1)	Before open: "Submissions open {date}"	n/a
Judges	Judge list with invite status and load	Name, organisation, assigned count	Invite, resend, deactivate	"Invite judges to start assigning"	One role per event check

Judge assignment	Matrix or list per round; required vs assigned counts; conflict flags	Teams × judges	Assign, bulk assign, unassign, auto-balance (P1), resolve conflict	No judges or no teams: explains which is missing	Unassign confirmation when submitted
Rubric builder	Template picker, criteria table, live weight total, state	Weights, ranges, descriptions	Add/edit/reorder criteria, activate, unlock	Template picker shown first	Structural edits disabled when Locked
Evaluation monitoring	Per judge and per team progress; missing list; correction requests	Counts, statuses	Open/close judging, reassign, exclude, approve/decline correction, remind (P1)	Judging not open: preconditions	n/a
Results review	Ranking per scope, flags, breakdown drawer, approval checklist, snapshot history	T, ranks, per-judge J, exclusions	Resolve tie, mark not ranked, approve, publish, unpublish, shortlist (P1)	"Results appear after judging opens"	Approve/publish confirmation dialogs

Feedback management	Form editors (participant, judge), responses summary, comment release	Response rate, aggregates	Edit, open, close, export, hide comment, release comments (P1)	"No responses yet"	Confidential forms never show identity
Certificate management	Templates per type with placement editor and preview; generation job; review list	Status counts, flags	Upload background, place fields, preview, generate, review, release, revoke, approve corrections	"Create a template to begin"	Winner generation needs published results
Staff tasks and issues	Task list, task form, issue queue	Status, assignees, severity	Create/edit task, acknowledge/resolve issue	"No tasks" / "No issues reported"	n/a
Integration settings	Sheets and Drive cards, job history, errors	Status, last success, last attempt	Connect, Sync now, retry, backfill, disconnect	"Not connected – use CSV export meanwhile"	Personal-data acknowledgement on connect
Exports and audit history	Export list; audit table with filters	Entity, actor, time, before/after	Export CSV, filter audit	"No changes recorded yet"	Read-only audit

15.3 Participant

Screen	Purpose and main components	Actions	Empty / special states	Permission notes
Event landing page	Public event info, tracks, guidebook link, primary action	Register / go to team / read guidebook	Registration not open / closed messages	Public
Registration	Fixed + custom fields, declarations, privacy notice	Submit	Already registered redirect; closed	Requires sign-in
Team creation / joining	Two options: create (name) or join (code) with confirmation	Create, join	Full team, invalid code, rate limit messages	Disabled after registration close
Team dashboard	Status, deadlines, submission status, announcements, results, certificates	Navigate to tasks	Incomplete-team warning; draft-not-submitted warning	Own team only
Team page	Members, Team ID, join code, leader controls	Remove, transfer, leave, regenerate code, rename	Locked after close	Leader-only actions hidden for members and rejected by server
Guidebook	Sections, updated date, PDF download	Read, download	Not published placeholder; PDF unavailable message	Published only

Track / problem selection	Options with descriptions	Select, change before lock	Locked message after Round 1 deadline	Leader only
Submission form	Requirements, fields, upload progress, draft state, countdown	Save draft (automatic), upload, submit, resubmit, discard draft	Before open: "Opens {date}"; locked; late-policy message	Own team only
Submission status	Current status, version history, receipt	View version	"Not started"	n/a
Results and team feedback	Winners list, own rank if enabled, released comments	Read	Not published placeholder	Own team comments only
Feedback form	Questions, confidentiality notice	Submit once	Not open / closed / already submitted	Eligible participants
Certificates	Event certificates list	Download, request name correction	Not released; revoked message	Own only
My profile	Registration fields	Edit allowed fields	Locked fields show "Contact organisers"	Own only

15.4 Judge

Screen	Purpose and main components	Actions	Empty / special states	Permission notes
Assigned rounds	Rounds with judging status, deadline, progress	Open round	"No teams assigned yet"	Assigned rounds only
Assigned teams	Table with submission and evaluation status, search, filters	Open team, filter	"All evaluations submitted" when done; Did-not-submit teams greyed	Assigned teams only
Team evaluation workspace	Three panes: team info, submission viewer, rubric form; next/previous	Declare conflict, report missing file/link, score, comment, submit, request correction	Judging not open: read-only; file load error with report; autosave status	Assignment checked on every request
Submission viewer	In-page PDF viewer, file list, link list, version label	Open, download, open links	File error state with report	Signed links only
Rubric scoring form	Criteria with descriptions, weights, ranges; live weighted score; comments	Enter scores, save (automatic), submit	Validation errors inline	Cannot alter rubric
Evaluation confirmation	Summary of submitted scores, timestamp	Next team, back to list	n/a	Read-only

Judge feedback form	Event feedback questions	Submit once	Not open / already submitted	Active judges

15.5 Helping Staff

Screen	Purpose and main components	Actions	Empty / special states	Permission notes
Assigned tasks	Task list with status, due time, location	Open task, filter	"No tasks assigned yet"	Own tasks only
Task details	Description, linked teams, status control	Change status, add note, open linked team	Blocked requires note	Assigned only
Check-in / verification	Search box, team card with members, per-member present and ID-verified toggles	Search, mark present, verify, undo, report issue	No match message; offline toggle revert	Limited fields only; no email
Task status update	Inline status change with confirmation on Done	Update	n/a	Assigned only
Issue reporting	Category, severity, team, description; list of own issues with status	Submit, escalate	"No issues reported"	Own reports only

16. Acceptance criteria and test scenarios

These are the release-gating tests; each maps to the requirement IDs shown. Example data uses Event "HackSonipat 2026" (team size 2–4, deadline 15 Feb 2026 18:00 IST, grace 0, late policy Reject unless stated).

ID	Workflow	Given	When	Then
AT-01	Create and publish event (FR-003, FR-005)	An admin has a Draft event with team size set and Round 1 deadline set	They click Publish	Status becomes Published; the public link shows the event; an audit entry records the publish
AT-01 N	Publish blocked	A Draft event with no round deadline	Admin clicks Publish	Publish fails listing "Round 1 submission deadline"; status stays Draft
AT-02	Register and create team (FR-051, FR-052)	Registration is Open; Asha is signed in and not registered	She completes the form, ticks declarations, creates team "Byte Me"	One registration exists; team gets code T-0001 and a join code; Asha is leader; admin list shows the team as Incomplete (1/2 minimum)
AT-02 N	Duplicate registration	Asha is registered	She submits the registration form again from another tab	No second registration; she is redirected to the team dashboard
AT-03	Join existing team (FR-053)	Team "Byte Me" has 3 of 4 members	Ravi enters the valid join code and confirms	Ravi becomes a member; team shows 4/4 and Complete
AT-03 N1	Join full team	Team has 4 of 4 members	Meena enters the join code	Join fails with "Team Byte Me is full

				(4/4)…"; no membership created
AT-03 N2	Simultaneous last slot	Team has 3 of 4; Meena and Kiran submit joins at the same moment	Both requests are processed	Exactly one succeeds; the other receives the full-team message; team has 4 members
AT-03 N3	Second team	Ravi is in "Byte Me"	He enters another team's code	Join fails naming "Byte Me"; membership unchanged
AT-04	Update participant details (FR-056, FR-019)	Registration is closed; Asha's name on certificate is "Asha K"	Asha changes it to "Asha Kumari"	Saved; admin detail and next Sheets push show the new value; audit records before/after
AT-04 N	Locked field	Registration is closed	Asha tries to change institution	Field is read-only; API rejects the change
AT-05	Upload submission (FR-065, FR-066)	Round 1 is open; form requires title, summary, PDF deck, repo URL	A member uploads a 12 MB PDF, enters a valid github.com/owner/repo URL and submits at 17:40 IST	Version 1 is Submitted at 17:40 IST by that member; all members see it; receipt email sent
AT-05 N1	Wrong file type	Same form	A member uploads deck.pdf that is actually a PPTX	Upload rejected with type error; draft remains without file

AT-05 N2	Incomplete	Repo URL empty	Member clicks Submit	Submit fails listing "Repository URL"; draft retained
AT-06	Edit before deadline (FR-072)	Version 1 submitted at 17:40	A member replaces the deck at 17:55 and clicks Resubmit	Version 2 is official; version 1 remains viewable to admins; status Submitted at 17:55
AT-06 N	Unsubmitted edit at deadline	Version 1 submitted; a member edits at 17:58 but does not resubmit	Deadline passes	Version 1 is locked as official; the draft is not evaluated
AT-07	Block late submission (FR-071)	Late policy Reject, grace 0	A team submits at 18:00:01 IST	Request rejected; message shows the deadline; no version created
AT-07 B	Late accepted	Late policy Accept and flag	A team submits at 18:07	Version accepted with status "Late by 7 min"; admin list flags it
AT-08	Sheets sync (FR-172–FR-174)	Sheets connected; 40 participants exist	5 new participants register within one minute	Within 5 minutes the Participants tab has exactly 45 data rows, no duplicates, and "Last synced" updates

AT-08B	Repeated sync	Same state	Admin clicks Sync now three times	Tab content identical; no duplicate rows
AT-09	Recover failed sync (FR-175–FR-178)	Google access is revoked from the Google account	A registration triggers a sync	Status shows "Google access was removed. Reconnect…"; last successful sync time unchanged; registration itself succeeded; after reconnecting, the next sync writes all current rows
AT-10	Assign judges (FR-027)	Round 1 has 30 teams and 6 judges; required evaluations = 2	Admin bulk-assigns so each team has 2 judges	Each team shows 2/2; each judge sees only their assigned teams
AT-11	Configure rubric (FR-023, FR-024)	Round 1 has no rubric	Admin selects Hackathon default and activates	Rubric has 5 criteria, weights total 100, state Active
AT-12	Validate weights (FR-025, FR-143)	Custom rubric with weights 30, 30, 30	Admin clicks Activate	Activation fails: "Weights total 90%; they must total 100%"
AT-12N	Invalid range	Criterion min 1, max 1	Admin saves and activates	Activation fails with "Max must be greater than min"

AT-13	Submit evaluation (FR-098)	Judging Open; judge assigned to T-0007; rubric 3 criteria	Judge scores all criteria, adds comment, confirms submit	Evaluation Submitted with server time; read-only for judge; progress increases by 1
AT-13 N1	Incomplete evaluation	Same	Judge leaves one criterion blank and submits	Fails listing the blank criterion; status stays Draft
AT-13 N2	Double submit	Same	Judge double-clicks Submit	One evaluation, one audit entry
AT-13 N3	Before judging opens	Deadline not yet passed	Judge opens T-0007	Scoring disabled with "Judging opens after submissions close"
AT-14	Weighted score (FR-146, FR-147)	Rubric A 30% (1–10), B 40% (1–10), C 30% (0–5); Judge 1 scores 8/7/4, Judge 2 scores 6/9/3	Both submit	J1 = 76.0000, J2 = 72.0000, team T = 74.0000
AT-14 N	Invalid score via API	Same rubric	A request sends B = 11	Rejected; no evaluation change
AT-15	Unauthorised judge access (FR-102)	Judge X is not assigned to T-0009	X requests T-0009's submission or evaluation by ID	Response is not found; denied access logged

AT-16	Approve and publish (FR-033–FR-035)	Judging Closed; all evaluations submitted; T-0003 and T-0011 tie at 3rd with 3 award positions	Admin attempts approve	Blocked until tie resolved; after "Share position" with reason, approval creates snapshot v1; Publish shows winners to participants
AT-16 N1	Missing evaluation	One assignment unsubmitted	Admin attempts approve	Blocked listing the assignment
AT-16 N2	Change after approval	Snapshot v1 approved	Admin reopens an evaluation	Approval invalidated; status Review; published view (if any) remains v1 until v2 is approved and published
AT-17	Collect feedback (FR-210–FR-214)	Participant form Open, Confidential	Asha submits	Response stored; Asha sees thank-you; admin sees aggregates and text without Asha's name; second submission blocked
AT-18	Generate and release certificates (FR-231–FR-235)	Final results published; participation template valid; 120 eligible participants	Admin generates, reviews, releases Participation	120 certificates with unique codes; generating again creates none; after release each participant downloads only their own

AT-18 N	Not released	Certificates generated, not released	Participant opens Certificates	Sees "Certificates will be available after the organisers release them"; no download URL exists
AT-19	Role isolation across events (Section 8.2)	Priya is Judge in Event A and Participant in Event B	Priya requests Event B's judge assignment list	Not found; her Event A judge role grants nothing in Event B
AT-19 N	Second role in same event	Priya is Participant in Event B	Admin invites her as Judge in Event B	Invite fails naming her existing role
AT-20	Restrict staff permissions (Module D)	Sahil is Staff	He requests a team's submission, an evaluation, the rubric, results before publication, or participant emails	Each returns not found; lookup cards contain no email
AT-21	Staff check-in (FR-126)	Team T-0042 has 4 members not checked in	Sahil marks all 4 present	Team shows Checked in; admin dashboard check-in count increases by 4; second tap does not duplicate
AT-22	Certificate name correction (FR-236)	Released certificate shows "Asha Kumri"	Asha requests correction to "Asha Kumari"; admin approves	Same certificate code, version 2, new name; old PDF no longer downloadable

AT-23	Rubric unlock (FR-153)	Rubric Locked; 10 evaluations submitted	Admin unlocks and changes a weight	Confirmation states 10 evaluations will reopen; after confirming, all 10 are Reopened and none are counted until resubmitted

17. Risks, dependencies, and open questions

The biggest risks are scope (too much for a first cycle), scoring correctness, and event-day reliability; integration risk is contained by making Google services optional.

17.1 Risk register

Probability and impact: L = low, M = medium, H = high.

ID	Category	Risk	Prob.	Impact	Mitigation	Owner
R-01	Product	Scope exceeds team capacity; nothing is pilot-ready	H	H	Build the smallest coherent version (13.3) first; P1 items only after Phase 4 exit	Product lead
R-02	Product	Pilot event's format differs from assumptions (e.g. live-pitch scheduling, multi-round)	M	H	Confirm pilot event structure before Phase 1 (OQ-03)	Product lead

R-03	Technical	Scoring bug produces wrong results	M	H	Formula test fixtures; independent spreadsheet recalculation; approval step; snapshots	Engineering (judging)
R-04	Technical	Deadline load spike slows or fails submissions	M	H	Load test at 150 submissions/10 min; direct-to-storage uploads if the stack supports it; admin-wide extension procedure	Engineering (platform)
R-05	Technical	Certificate rendering issues (long names, fonts, scripts)	M	M	Shrink-to-fit rule; test with real names; preview before release	Engineering (certificates)
R-06	Integration	Google OAuth app verification or quota limits delay Sheets/Drive	M	M	Early spike in Phase 2; per-file scope; integrations optional; CSV fallback	Engineering (integrations)
R-07	Integration	Token revoked or sheet deleted mid-event	M	L	Truthful status, retries, reconnect flow, CSV	Engineering (integrations)
R-08	Data consistency	Organisers edit the managed sheet expecting	H	M	Banner row, connection warning, help page; overwrite	Product + design

		changes to import			behaviour explained in pilot onboarding	
R-09	Data consistency	Late admin edits (membership, disqualification) after results approved go unnoticed	L	H	Automatic approval invalidation with cause shown	Engineering (judging)
R-10	Live operations	Venue Wi-Fi failure during check-in or judging	M	M	Autosave, idempotency, offline banner; printed/CSV fallback roster; mobile data hotspot	Event operations
R-11	Live operations	Judges unfamiliar with the tool slow down judging	M	M	5-minute judge briefing, test login before event day, demo team	Event operations
R-12	Live operations	No developer available during event if something breaks	M	H	Named on-call developer; event-day runbook	Engineering lead
R-13	Privacy / access	Permission bug exposes scores or personal data	M	H	Server-side checks, matrix test suite, not-found responses, code review of every endpoint	Engineering (platform)

R-14	Privacy	Sheets shared widely by organisers leaks phone numbers	M	M	No scores in Sheets; acknowledgement at connect; option to omit phone column (P1)	Product
R-15	Adoption	Organisers keep running parallel spreadsheets out of habit	M	M	Ensure CSV and Sheets cover their needs; observe and interview (H-1)	Product lead
R-16	Adoption	Students resist creating accounts	L	M	Email sign-in link option; minimal form	Design
R-17	Fairness	Teams judged by different judges get unfair scores due to judge harshness	M	M	Assign balanced panels; judge spread view (P1); validate need for normalisation (H-5)	Product + organisers

17.2 Dependencies

- Technology stack, hosting, file storage and transactional email provider chosen (OQ-01, OQ-08).
- A confirmed pilot event with a date, organiser contact and format (OQ-03).
- A Google Cloud project and OAuth consent configuration for P1 integrations.
- Certificate background designs and signatures supplied by organisers.
- Institution approval of the privacy notice and retention period (OQ-15).

17.3 Open questions register

Defaults are what this PRD assumes until the question is closed.

1. Decisions required before development

ID	Question	Default assumed in this PRD	Decider
OQ-01	Which technology stack, hosting and storage?	None assumed	Engineering lead
OQ-02	Who may create events during the pilot: any verified user, or an allowlist?	Any verified user; allowlist available operationally	Product lead
OQ-03	What is the pilot event (type, date, size, rounds, in-person or online, live pitches)?	Single-round hackathon, in person, ≤ AS-01 size	Product lead + organisers
OQ-04	Should complete drafts be auto-submitted at the deadline?	No auto-submit; prominent warnings	Product lead + organisers
OQ-05	What should teams see when results publish: winners only, own rank, own score?	Winners list on; own rank and scores off	Organisers
OQ-06	Default ranking scope (overall vs per track) and number of award positions?	Per track when tracks exist; 3 award positions	Organisers
OQ-07	Can a participant's login email be changed, and by whom?	Not in V1	Product lead
OQ-08	Which email provider, and what sending limits apply?	Transactional email for auth, invites, receipts only	Engineering lead

2. Decisions that can be made during implementation

ID	Question	Default assumed	Decider
OQ-09	Participation certificate rule: submitted team, or checked in?	Submitted in Round 1	Organisers
OQ-11	How to preview PPT/PPTX inline for judges (conversion or download only)?	Download only in V1	Engineering
OQ-12	Can admins upload a submission on a team's behalf (e.g. emailed file during an outage)?	No; admin extends the deadline instead	Product lead
OQ-13	Should a judge also be able to act as staff in the same event?	No	Product lead
OQ-14	Final session timeouts per role	12 h admin, 7 days others	Engineering
OQ-15	Data protection obligations and retention period for the institution (must close before pilot launch)	12 months after conclusion	Faculty coordinator / institution
OQ-16	Must certificates support non-Latin scripts (e.g. Devanagari names)?	Latin only unless confirmed	Organisers

3. Decisions that can wait until after the pilot

ID	Question	Evidence needed

OQ-10	Do multi-round events need cumulative scores across rounds?	Organiser interviews (H-4)
OQ-17	Is anonymous judging needed?	Case competition organisers (H-6)
OQ-18	Is judge-bias normalisation needed?	Pilot judge spread data (H-5)
OQ-19	Is bidirectional Sheets sync worth its cost?	Pilot observation (H-1, H-2)
OQ-20	Multiple files per submission field?	Submission patterns in pilot
OQ-21	Public certificate verification page?	Participant and employer requests
OQ-22	Which notifications (reminders, email copies of announcements) add most value?	Pilot feedback

18. Final product recommendations

The proposed scope is too large to build in full for a first student release; build the smallest coherent version first, then add integrations, and run the pilot with CSV and native storage if integrations are not ready.

18.1 Recommended MVP scope

The smallest coherent version in 13.3, plus staff check-in and tasks for an in-person pilot, plus judge feedback form. All P0 rows in Section 6.

18.2 Remove or simplify

Item	Recommendation
Google Drive as primary storage	Replace with native storage; Drive becomes an optional archive mirror
Google Sheets sync	One-way snapshot push only; CSV export is the guaranteed path
Staff module	Tasks with one status, check-in and issues; no task board, no queue management
Feedback	Two fixed forms from defaults; comment release P1
Rubrics	All criteria required; integers only; no normalisation; no optional criteria
Multi-round	Data model supports it; shortlist UI is P1
Certificates	Background image plus placed fields; no drag-and-drop designer; verification page P2
Notifications	Only auth, invites and submission receipts by email

18.3 Five highest-risk decisions

1. Scoring formula and completeness rules (E.1, FR-149): any ambiguity becomes a public dispute.
2. Deadline semantics (server time, grace, drafts not auto-submitted, OQ-04).
3. One role per user per event (8.2): simple and safe, but may frustrate small teams who double up.
4. Platform as source of truth with one-way Sheets push: organisers must accept that sheet edits are overwritten.

5. Native storage over Drive: simpler access control, but organisers lose the familiar Drive-first workflow.

18.4 Five most important user journeys

1. Participant registers, forms a team and submits before the deadline (7.B).
2. Judge evaluates all assigned teams (7.C).
3. Organiser reviews, approves and publishes results (7.A, A16–A18).
4. Organiser sets up and publishes the event with its guidebook (7.A, A1–A9).
5. Participant downloads a correct certificate (7.B, B16; Module I).

18.5 Recommended implementation order

Phase 1 foundations → Phase 2 content and submissions (with OAuth spike) → Phase 3 judging and results → Phase 4 event-day operations, feedback and certificates → Phase 5 integrations and hardening (Section 14).

18.6 Minimum requirements for a successful pilot

- Every submission is associated with the correct team, and the official version is unambiguous.
- Results are calculated, approved and published in the platform, and an independent recalculation matches.
- No permission breach: judges, staff and participants saw only what Section 8 allows.
- Participants downloaded certificates from their accounts.
- Pilot metrics in Section 12 were captured, including baselines.

18.7 Assumptions to validate with organisers first

H-1 (platform as source of truth), H-4 (round structure), H-5 (plain mean is acceptable), H-6 (no blind judging), OQ-04 (no auto-submit), and OQ-05 (what teams see in results).

18.8 Launch-readiness checklist

- Pilot event configuration reviewed and signed off by the organisers
- All P0 acceptance tests in Section 16 pass
- Permission test suite covers every matrix cell and passes
- Scoring results for a seeded event match an independent spreadsheet
- Load test at the Section 11 profile passes the deadline scenario
- Backup taken and a restore rehearsed
- Privacy notice and retention period approved (OQ-15)
- Certificate templates previewed with the longest real names
- Judges and staff have logged in at least once before event day
- CSV exports of teams and check-in printed or saved as offline fallback

- Event-day runbook written: extension procedure, outage procedure, contacts
- On-call developer named for the event window
- Integrations either verified with fault tests or switched off with CSV fallback communicated
- Observer assigned to record manual metrics during the event

18.9 Consistency check

The pairs the brief asked to check, with how this PRD resolves each.

Area	Potential contradiction	Resolution
Permissions vs features	Admins manage submissions but cannot upload or edit them	Admins extend or unlock; uploading on a team's behalf is OQ-12
Submission rules vs deadlines	Judges could score work that is still changing	Judging opens only after deadline + grace (FR-029); a later team extension pauses that team's assignments and reopens submitted evaluations on resubmission (7.C)
Submission rules vs team rules	Track can change until the Round 1 deadline, after judges may be assigned by track	Change allowed with an admin warning to review assignments (FR-061, edge case 27)
Rubric vs calculation	Editing a rubric mid-judging would mix versions	Rubric locks at first submitted evaluation; structural change requires unlock that reopens all evaluations (FR-153)
Judge assignment vs evaluation visibility	Removed or deactivated judges	Unassigned → evaluation excluded (FR-155); deactivated → submitted evaluations still counted unless an admin

		excludes them (8.4); access ends on the next request either way
Application vs Google Sheets	Organisers may expect sheet edits to flow back	One-way only, stated in UI and sheet banner; nothing is imported (6-F)
Storage vs Drive permissions	Judges might lack Drive access	Judges never use Drive; files come from native storage via signed links (FR-195)
Approval vs publication	Publishing stale or unapproved results	Only an approved snapshot can be published; later changes invalidate approval while the last published version stays visible until republished (FR-034, FR-035)
Certificate eligibility vs release	Disqualification or membership change after generation	Regeneration flags ineligible holders; admin revokes; winner certificates require published final results (FR-231, FR-232, FR-237)
Feedback anonymity vs duplicate prevention	"Anonymous" forms that store identity	Called "Confidential", not anonymous; identity stored only for duplicate prevention and never shown (FR-213)